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FATİH SULTAN MEHMET VAKIF ÜNİVERSİTESİ**

Computer Engineering

**BRAIN TUMOR CLASSIFICATION AND SEGMENTATION
SYSTEM USING DEEP LEARNING
Project Report**

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1 FOREWORD

This graduation project, conducted under the Computer Engineering Department at Fatih Sultan Mehmet Vakıf University, Faculty of Engineering, represents the culmination of two semesters of research and development in the field of medical image analysis using deep learning techniques.

The project evolved significantly from its initial conception as a binary classification system (tumor vs. no tumor) in the first semester (Spring 2025) to a comprehensive multi-class classification and segmentation system in the second semester (Fall 2025). The final deliverable includes a custom Convolutional Neural Network (CNN) capable of classifying brain MRI scans into four categories (Glioma, Meningioma, Pituitary Tumor, and No Tumor) with **99.03% accuracy**, a U-Net-based segmentation model for tumor localization, and a fully functional web application for clinical use.

The classification system was developed using custom-designed convolutional neural networks built from scratch without the use of pretrained models, trained on a comprehensive dataset of over 43,000 MRI images aggregated from multiple publicly available sources. The segmentation model was trained on the LGG Brain Segmentation dataset from Kaggle, enabling precise tumor boundary delineation.

We would like to express our sincere gratitude to our advisor, **Dr. Öğr. Üyesi Abdulkерim Mohammed YİBRE**, for his continuous guidance, valuable insights, and unwavering support throughout every phase of this project.

With respect,

Nurettin Süleymanoğlu, Muhammed Emin Korkunc, Abdullah Sina Korkmaz

2 ABSTRACT

Brain tumors are abnormal growths of cells within the brain that pose serious health risks and can be life-threatening if not detected early. While Magnetic Resonance Imaging (MRI) is the standard modality for brain tumor detection, manual interpretation is time-consuming and prone to human error. This graduation project presents the development of a comprehensive Deep Learning-Based Brain Tumor Classification and Segmentation System designed to assist radiologists in the diagnostic process.

The project was developed in two phases. **Phase I** (Spring 2025) focused on binary classification, achieving **97.21% accuracy** with **99.05% precision** and **94.33% recall** on a dataset of approximately 7,000 MRI images. **Phase II** (Fall 2025) expanded the system to multi-class classification across four categories (Glioma, Meningioma, Pituitary, No Tumor) using an aggregated dataset of **43,687 images**, initially achieving **95.09% accuracy** and ultimately reaching **99.03% accuracy** through architectural refinements and optimization techniques.

Additionally, a U-Net-based segmentation model was implemented to provide pixel-wise tumor localization, achieving a validation loss of **0.1253** with a Dice coefficient of approximately **0.87**. The complete system was deployed as a web application called **NeuroScan AI**, featuring a React frontend and Flask backend, enabling real-time MRI analysis with classification results, confidence scores, and visual segmentation masks.

The system demonstrates exceptional reliability, achieving **100% recall for healthy patients** (No Tumor class) and per-class F1 scores exceeding **98%** for all tumor categories. This work contributes to the growing body of evidence that AI-powered diagnostic tools can significantly enhance medical imaging workflows and support clinical decision-making.

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5 LIST OF ABBREVIATIONS

Abbreviation	Definition
AI	Artificial Intelligence
AUC	Area Under the Curve
BCE	Binary Cross-Entropy
BraTS	Brain Tumor Segmentation Challenge
CAD	Computer-Aided Diagnosis
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
IoU	Intersection over Union
LGG	Low-Grade Glioma
MRI	Magnetic Resonance Imaging
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
SVM	Support Vector Machine
TN	True Negative
TP	True Positive

6 INTRODUCTION

6.1 Background and Motivation

Brain tumors are abnormal growths of cells within the brain that pose serious health risks and can be life-threatening if not detected early. According to the World Health Organization, brain tumors account for approximately 2% of all cancers, yet they are responsible for a disproportionately high rate of cancer-related mortality due to the critical nature of the affected organ [1]. In clinical practice, Magnetic Resonance Imaging (MRI) is the preferred modality for brain tumor detection and diagnosis because it provides high-resolution images of soft tissues without ionizing radiation [2].

However, interpreting MRI scans for tumor detection is a challenging and time-consuming task for radiologists. Manual diagnosis from MRI slices can be error-prone due to subtle differences between tumor and normal tissue, the variability in tumor appearance across patients, and the sheer volume of images that need to be analyzed in modern clinical settings [3]. Early and accurate detection of brain tumors is crucial for improving patient survival rates, yet traditional biopsy and histopathological methods are invasive, slow, and subject to sampling errors [4]. These challenges highlight the urgent need for computer-assisted diagnostic tools to support radiologists in identifying tumors more efficiently and consistently.

In recent years, artificial intelligence (AI)—and deep learning in particular—has emerged as a powerful approach to assist in medical image analysis. Convolutional Neural Networks (CNNs), a class of deep learning models specialized for image data, have demonstrated remarkable success in automatically learning discriminative features from images and outperforming earlier hand-crafted feature methods in tasks like tumor classification [5]. The ability of CNNs to learn hierarchical features directly from raw MRI scans makes them well-suited for detecting the complex patterns of brain tumors without requiring manual feature engineering [6].

Deep learning-based techniques have already achieved high accuracy in identifying tumors in brain MRI, often matching or exceeding human expert performance on specific benchmarks [7]. This has motivated extensive research into deep learning models for brain tumor detection and classification. Building on these advances, this project aims to develop a comprehensive Deep Learning-Based Brain Tumor Classification and Segmentation System that can automatically determine whether an MRI scan contains a brain tumor, classify the tumor type, and localize its boundaries.

6.2 Problem Statement

The diagnosis of brain tumors from MRI scans presents several significant challenges:

1. **Diagnostic Complexity:** Different tumor types (Glioma, Meningioma, Pituitary) require vastly different treatment approaches. A binary “tumor present” vs. “no tumor” classification has limited clinical utility because it does not inform treatment planning [8].
2. **Volume of Data:** Modern MRI machines produce hundreds of slices per patient scan, making manual review extremely time-consuming. Radiologists face increasing workloads, which can lead to fatigue and diagnostic errors [9].

3. **Subtle Visual Differences:** Early-stage tumors may have intensity profiles very similar to healthy tissue, making them difficult to detect even for experienced radiologists [10].
4. **Need for Localization:** Beyond detecting the presence of a tumor, clinicians need to know the exact location and extent of the tumor for surgical planning and treatment monitoring [11].
5. **Accessibility:** In many regions, particularly in developing countries and rural areas, access to specialized neuroradiologists is limited. Automated diagnostic tools could democratize access to expert-level analysis [12].

These challenges motivate the development of an automated system that can:

- Classify MRI scans into multiple tumor categories with high accuracy
- Achieve sensitivity levels that minimize missed diagnoses (false negatives)
- Provide visual localization of tumor regions
- Operate in real-time through an accessible interface

6.3 Project Objectives

This graduation project was designed with the following specific objectives:

Phase I Objectives (Spring 2025):

1. Compile a diverse dataset from multiple publicly available MRI brain scan sources
2. Design and implement a CNN-based binary classifier (tumor vs. no tumor)
3. Achieve classification accuracy exceeding 95% on held-out test data
4. Evaluate model performance using precision, recall, F1-score, and confusion matrix analysis

Phase II Objectives (Fall 2025):

1. Expand the system to multi-class classification (Glioma, Meningioma, Pituitary, No Tumor)
2. Aggregate a large-scale dataset (>40,000 images) to support complex learning
3. Achieve multi-class classification accuracy approaching (>98%)
4. Implement a U-Net-based segmentation model for tumor localization
5. Develop a web-based application for clinical deployment
6. Create a complete end-to-end Computer-Aided Diagnosis (CAD) pipeline

6.4 Project Scope

In Scope:

- Binary classification of brain MRI images (Phase I)
- Multi-class classification into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor (Phase II)
- Pixel-wise tumor segmentation using U-Net architecture
- Development of a web application with image upload, classification, and segmentation visualization capabilities
- Comprehensive evaluation using standard machine learning metrics

Out of Scope:

- 3D volumetric analysis (the system processes 2D MRI slices)
- Multi-modal MRI fusion (T1, T2, FLAIR combined analysis)
- Real-time video stream processing

- Integration with hospital Picture Archiving and Communication Systems
- Regulatory approval for clinical deployment

Datasets Used:

- **For Classification:** Combined Multi-Class dataset from five sources totaling 43,687 images [13-17]
- **For Segmentation:** LGG Brain MRI Segmentation dataset from Kaggle [18]

6.5 Report Organization

This report is organized as follows:

Chapter 2: Literature Review and Related Work provides an overview of traditional methods and deep learning advances in brain tumor detection, discusses relevant CNN architectures, and reviews the state-of-the-art in medical image segmentation.

Chapter 3: Phase I - Binary Classification System details the first semester's work, including dataset preparation, model architecture, training methodology, and evaluation results achieving 97.21% accuracy.

Chapter 4: Phase II - Multi-Class Classification System describes the evolution to four-class classification, dataset expansion to 43,687 images, and the progression from 95.09% to 99.03% accuracy through model refinement.

Chapter 5: Tumor Segmentation with U-Net presents the implementation of the segmentation model, including architecture design, training strategy, and results.

Chapter 6: Web Application - NeuroScan AI documents the development of the complete web-based system, including backend API, frontend interface, and deployment architecture.

Chapter 7: Discussion analyzes the clinical significance of the results, compares performance with related work, and discusses limitations.

Chapter 8: Conclusion and Future Work summarizes the achievements and outlines potential extensions and improvements.

7 LITERATURE REVIEW AND RELATED WORK

Research on automated brain tumor detection from MRI has evolved significantly over the past decade. Early approaches relied on conventional machine learning methods with manually engineered features, while recent work leverages deep learning for superior performance. This chapter reviews the progression of techniques and establishes the context for our approach.

7.1 Traditional Methods in Brain Tumor Detection

Before the deep learning revolution, researchers commonly used algorithms like Support Vector Machines (SVM), Random Forests, k-Nearest Neighbors, and other classifiers on hand-crafted features extracted from MRI images [19]. For example, texture analysis using Gray-Level Co-occurrence Matrices (GLCM) and wavelet transforms were employed to characterize tumor appearance, followed by SVM to classify tumor vs. non-tumor tissues [20].

Anwar et al. (2018) provided a comprehensive review of medical image analysis techniques prior to the widespread adoption of deep learning, documenting how feature engineering was essential for achieving reasonable classification accuracy [21]. These traditional methods achieved accuracy rates typically in the range of 80-90%, but had notable limitations:

- **Domain Expertise Requirement:** Significant expertise was needed to design effective features for tumor characterization
- **Limited Generalization:** Hand-crafted features often struggled with the variability in tumor shape, size, and intensity patterns across different patients and imaging protocols
- **Performance Plateau:** Classification performance plateaued on complex, high-dimensional image data when compared to newer approaches
- **Time-Intensive Development:** Feature engineering was a time-consuming iterative process

The introduction of deep Convolutional Neural Networks marked a paradigm shift, as these models could automatically learn the most relevant features directly from raw image data.

7.2 Deep Learning Advances in Medical Imaging

The introduction of deep CNNs has dramatically improved the state-of-the-art in brain tumor image analysis. CNNs can automatically learn complex feature hierarchies from the data, which has proven advantageous for capturing tumor characteristics that vary across patients [22].

Pereira et al. (2016) were among the early adopters of deep CNNs for brain tumor analysis, achieving notable improvements in tumor segmentation on the BraTS challenge dataset. Their work demonstrated that CNNs could capture hierarchical features—from low-level edges to high-level semantic structures—that are critical for distinguishing tumor tissue from healthy brain matter [23].

Swati et al. (2019) applied transfer learning with pre-trained deep CNN models (VGG16 and VGG19) on a dataset of T1-weighted MRI slices with three tumor types. They achieved approximately **94.82% accuracy** in multi-class tumor classification, demonstrating that even with minimal preprocessing and no handcrafted features, deep models could excel in this task [24].

Kumar et al. (2018) explored different CNN architectures (AlexNet, ResNet50, InceptionV3) for classifying primary tumor types and reported that AlexNet reached **96.2% accuracy**, sometimes outperforming deeper models in specific experimental setups [25]. This finding highlighted that the optimal architecture depends on dataset characteristics and that deeper is not always better.

Rasheed et al. (2023) presented a CNN-based classification system combining the Figshare, SARTAJ, and Br35H datasets (7,023 images) with preprocessing techniques including CLAHE (Contrast Limited Adaptive Histogram Equalization). They achieved approximately **97.84% accuracy**, demonstrating the importance of image enhancement in medical imaging tasks [26].

Abdusalomov et al. (2023) introduced a YOLOv7-based model with attention mechanisms for detecting and localizing brain tumors in MRI, achieving **99.5% accuracy**. This work represented the frontier of detection accuracy, though using a more complex architecture [27].

7.3 CNN Architectures for Image Classification

Several CNN architectures have become foundational in image recognition and have been extensively applied to medical imaging. Understanding these architectures informed our model design:

7.3.1 VGG (VGG16/VGG19)

The VGG network, introduced by Simonyan and Zisserman (2015), is characterized by its simple and uniform architecture using stacked 3×3 convolutional layers and periodic pooling [28]. VGG16 has 16 weighted layers (13 convolutional + 3 fully connected), while VGG19 has 19 layers.

Key Design Principles:

- Small (3×3) convolutional filters throughout
- Doubling of feature channels after each pooling operation
- Multiple consecutive convolutions before each pooling layer

The VGG architecture achieved top performance in the 2014 ImageNet competition and became a popular baseline for transfer learning in medical imaging. Many studies, including Swati et al. [24], used pre-trained VGG models for brain tumor classification with fine-tuning.

Our Approach: Our custom CNN architecture draws from VGG-style design with sequential convolutional blocks featuring increasing filter counts ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$), though with fewer total parameters to prevent overfitting on our dataset size.

7.3.2 ResNet (Residual Networks)

Introduced by He et al. (2016), ResNet revolutionized deep learning by enabling networks to go much deeper (50, 101, or even 152 layers) without suffering from vanishing gradients [29]. The key innovation is the **residual block**, where shortcut (skip) connections bypass convolutional layers and add the input directly to the output.

Impact on Medical Imaging:

- Raza et al. (2020) demonstrated that ResNet-152 outperformed VGG16 on MRI tumor classification [30]
- The ability to train very deep networks enabled learning of more complex and subtle features

7.3.3 DenseNet (Dense Connections)

DenseNet, introduced by Huang et al. (2017), connects each layer directly to all subsequent layers in a feed-forward fashion [31]. This dense connectivity ensures maximum information flow and feature reuse while alleviating vanishing gradient problems.

Advantages:

- Parameter efficiency through feature reuse
- Strong gradient flow during training
- Particularly effective when data is limited

DenseNet architectures (e.g., DenseNet-121) have been successfully applied to brain tumor classification, often outperforming other networks on smaller datasets [32].

7.4 U-Net and Biomedical Image Segmentation

The U-Net architecture, introduced by Ronneberger et al. (2015), has become the gold standard for biomedical image segmentation [33]. The architecture features an encoder-decoder structure with skip connections that enable precise localization.

Architecture Overview:

The U-Net architecture consists of an **encoder path** (contracting) and a **decoder path** (expanding) connected by skip connections:

- **Encoder Path:** Four convolutional blocks with feature channels $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$, each followed by max pooling for downsampling
- **Bottleneck:** Central layer with 1024 channels at the lowest spatial resolution
- **Decoder Path:** Four upsampling blocks using transposed convolutions, each concatenating with the corresponding encoder features via skip connections
- **Output:** 1×1 convolution producing the final segmentation mask

Key Features:

- **Skip Connections:** Concatenate feature maps from encoder to decoder, preserving spatial information
- **Symmetric Structure:** Encoder reduces spatial dimensions while decoder recovers them
- **Applicability to Small Datasets:** Originally designed to work with very limited training data through extensive augmentation

U-Net has achieved state-of-the-art results on the BraTS (Brain Tumor Segmentation) Challenge, with winning solutions often using U-Net variants [34]. The architecture's success in brain tumor segmentation motivated our choice for the segmentation component of this project.

7.5 Benchmark Datasets and State-of-the-Art Results

The field has been driven by publicly available datasets and challenges. Understanding these benchmarks contextualizes our work:

7.5.1 Figshare Brain Tumor Dataset

The Figshare dataset compiled by Cheng et al. contains **3,064 contrast-enhanced T1-weighted MRI images** with three tumor types: Glioma, Meningioma, and Pituitary tumor [35]. This dataset has been a de facto benchmark, with numerous studies reporting 93-96% accuracy on 3-class classification tasks.

7.5.2 Br35H Dataset

The Kaggle Br35H dataset contains approximately **3,000 T1-weighted MRI images** organized into “yes” (tumor) and “no” (no tumor) folders, making it suitable for binary classification tasks [36].

7.5.3 BraTS Challenge

The Brain Tumor Segmentation (BraTS) Challenge provides multi-modal MRI volumes with expert annotations for segmentation tasks. While focused on segmentation, techniques from BraTS have inspired classification models [37].

7.5.4 Summary of State-of-the-Art Results

Table 2.1: Comparison of Related Works in Brain Tumor Classification

Study	Year	Method	Dataset Size	Classes	Accuracy
Swati et al. [24]	2019	VGG19 Transfer Learning	3,064	3	94.82%
Kumar et al. [25]	2018	AlexNet	3,000	3	96.2%
Rasheed et al. [26]	2023	CNN + CLAHE	7,023	4	97.84%
Abdusalomov et al. [27]	2023	YOLOv7 + Attention	5,000	4	99.5%
Akbar et al. [38]	2022	ResNet + SVM Hybrid	3,000	3	98.0%
This Work (Phase I)	2025	Custom CNN	7,000	2	97.21%
This Work (Phase II)	2025	Custom CNN	43,687	4	99.03%

Our final model’s performance of **99.03% accuracy** places it among the top-performing systems in the literature, achieved through careful dataset curation, architecture design, and training optimization.

8 PHASE I: BINARY CLASSIFICATION SYSTEM

The first phase of this graduation project, conducted during Spring 2025, focused on developing a deep learning model to perform binary classification of brain MRI images—determining whether a tumor is present or not. This chapter details the dataset compilation, preprocessing methodology, model architecture, training process, and evaluation results.

8.1 Dataset Description and Compilation

Instead of relying on a single dataset, we compiled a custom dataset by combining multiple publicly available MRI brain scan datasets. This approach was chosen to maximize data diversity and improve model robustness across different imaging conditions.

8.1.1 Primary Data Sources

1. Kaggle Br35H Brain Tumor Detection Dataset [36]

This dataset consists of T1-weighted contrast-enhanced brain MRI images organized into two categories:

- **“Yes” (Tumor):** Approximately 1,500 images showing brains with visible tumors
- **“No” (No Tumor):** Approximately 1,500 images of healthy brain scans

All images were grayscale and provided at a uniform size of 224×224 pixels. The scans were collected from different patients and hospitals, ensuring variability in imaging conditions.

2. Hugging Face MRI Brain Tumor Dataset (by PranomVignesh) [17]

This dataset is an aggregate containing images of three tumor types and healthy brains:

- **Classes:** Glioma, Meningioma, Pituitary Tumor, and No Tumor
- **Total Images:** Approximately 5,370 images

- **Source Composition:** Compiled from the Figshare dataset (3,064 tumor images), SARTAJ dataset, and Br35H dataset

8.1.2 Combined Dataset Statistics

Table 3.1: Phase I Dataset

Composition

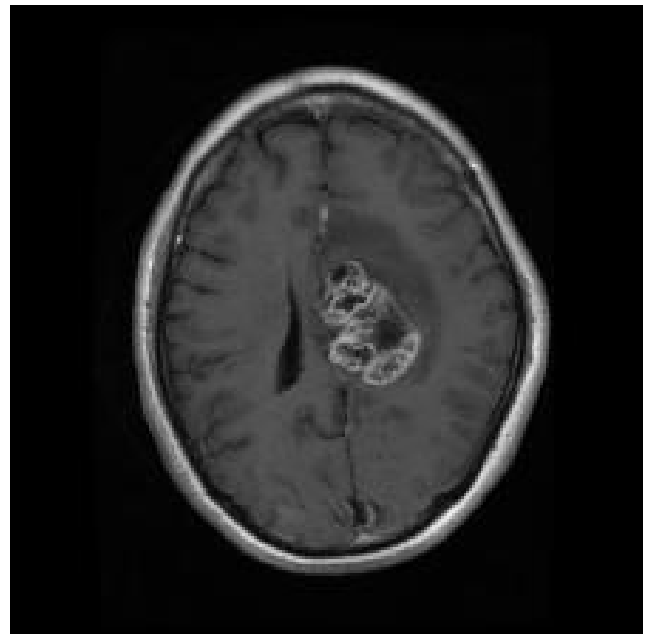
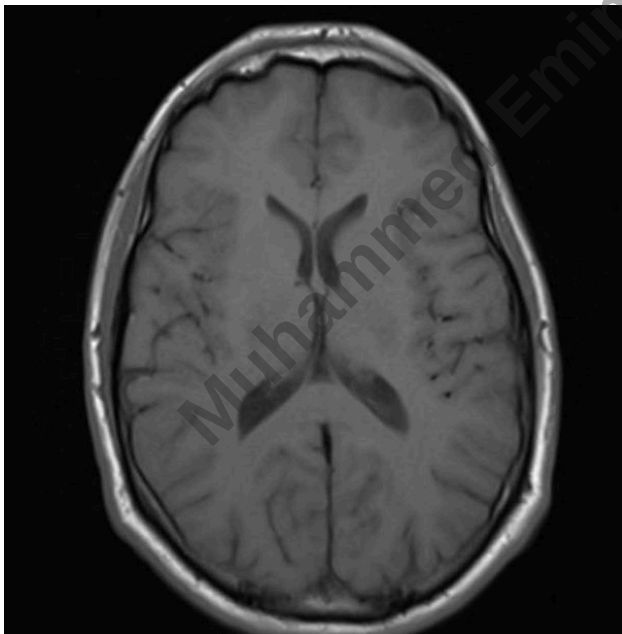
Category	Number of Images	Percentage
Tumor (Yes)	3,500	50%
No Tumor (No)	3,500	50%
Total	7,000	100%

For binary classification, all tumor subtypes (Glioma, Meningioma, Pituitary) were mapped to the single “Tumor” class (label = 1), while healthy brain images remained in the “No Tumor” class (label = 0).

8.1.3 Data Splitting Strategy

The combined dataset was split into three subsets using stratified sampling to maintain class balance:

- **Training Set:** 70% (4,900 images) — Used for model weight updates
- **Validation Set:** 15% (1,050 images) — Used for hyperparameter tuning and early stopping
- **Test Set:** 15% (1,050 images) — Held out completely for final evaluation



8.2 Data Preprocessing Pipeline

Raw medical images require preprocessing to normalize differences in acquisition and prepare data for the neural network. We implemented a comprehensive preprocessing pipeline:

8.2.1 Resolution Standardization

All images were resized to a fixed resolution of **256×256 pixels**. This size was chosen because:

- Many CNN architectures expect 224×224 or 256×256 inputs
- It provides a good balance between preserving anatomical detail and computational efficiency
- Resizing was performed using OpenCV's bilinear interpolation

8.2.2 Grayscale Handling

MRI scans are inherently grayscale images. Some datasets stored images in 3-channel RGB format (with identical channel values). We handled this by:

- Converting any RGB images to single-channel grayscale
- Duplicating the grayscale channel to 3 channels when required by the model architecture
- Final input shape: (3, 256, 256) per image

8.2.3 Intensity Normalization

MRI images can have varying intensity ranges due to different scanner settings. We applied two- step normalization:

Scaling: Pixel values were scaled from 0-255 to [0, 1] by dividing by 255.0

1. **Z-score Standardization:** Applied using dataset mean and standard deviation:
$$\text{normalized_pixel} = (\text{pixel} - \text{mean}) / \text{std}$$

This normalization helps the model converge faster and is especially important when comparing images from different sources.

8.2.4 Data Augmentation (Training Only)

To improve generalization and prevent overfitting, we applied real-time data augmentation to training images:

- **Random Horizontal Flip:** MRI axial slices are roughly symmetric left-right
- **Random Rotation:** ±15 degrees to account for slight head positioning differences
- **Random Zoom:** Scaling by up to ±10%
- **Intensity Jitter:** Small random adjustments to brightness and contrast
- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Applied selectively to improve local contrast and make tumor regions more discernible

Augmentation was applied only to training data and performed on-the-fly during each epoch, effectively increasing the dataset diversity without storing additional images.

8.3 Model Architecture

We designed a custom Convolutional Neural Network with five convolutional blocks followed by fully connected layers for classification. The architecture was influenced by VGG-style design principles but with reduced complexity appropriate for our dataset size.

8.3.1 Architecture Overview

The Phase I CNN architecture consists of two main components:

Convolutional Feature Extractor (5 Blocks):

- **Block 1:** Conv2D(1→16, 3×3) → BatchNorm → ReLU → MaxPool(2×2) — Output: 16×128×128

- **Block 2:** Conv2D(16→32, 3×3) → BatchNorm → ReLU → MaxPool(2×2) — Output: 32×64×64
- **Block 3:** Conv2D(32→64, 3×3) → BatchNorm → ReLU → MaxPool(2×2) — Output: 64×32×32
- **Block 4:** Conv2D(64→128, 3×3) → BatchNorm → ReLU → MaxPool(2×2) — Output: 128×16×16
- **Block 5:** Conv2D(128→256, 3×3) → BatchNorm → ReLU → MaxPool(2×2) — Output: 256×8×8

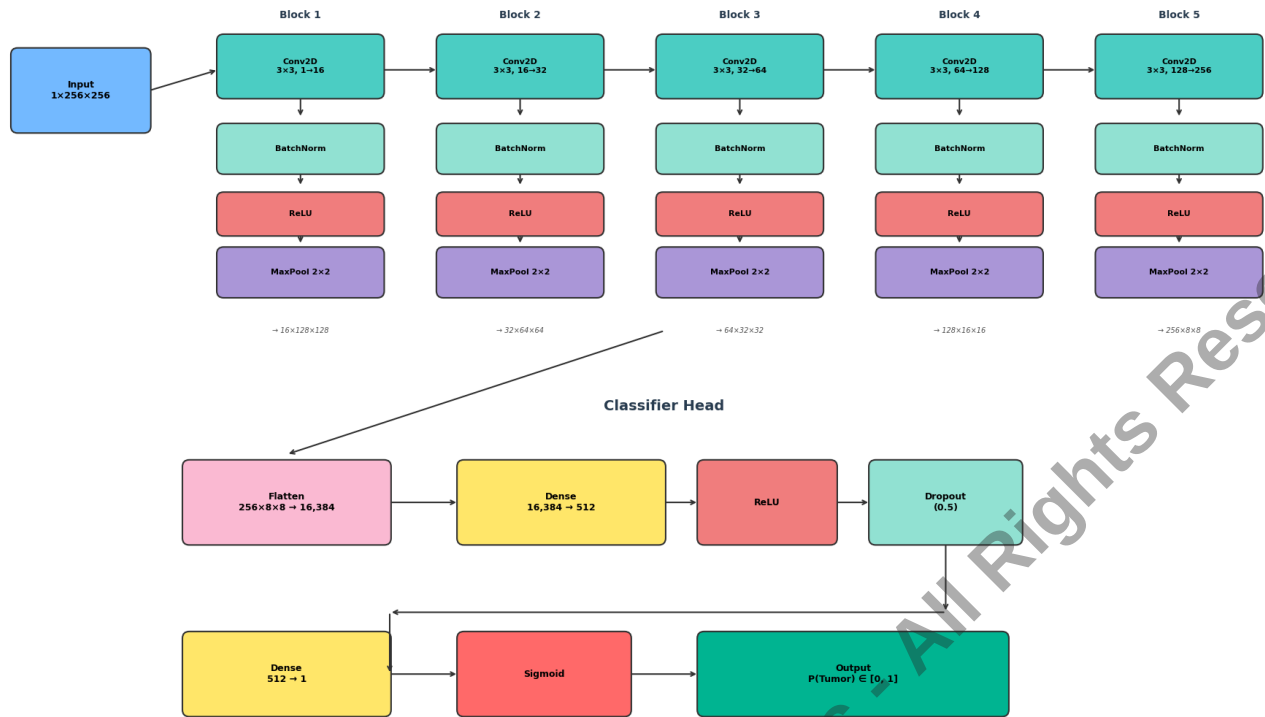
Classifier Head:

- Flatten: $256 \times 8 \times 8 = 16,384$ features
- Dense(16384→512) → ReLU → Dropout(0.5)
- Dense(512→1) → Sigmoid
- Output: Probability of tumor presence [0, 1]

8.3.2 Design Rationale

Component	Choice	Rationale
Filter Progression	16→32→64→128→256	Gradual increase captures increasingly complex features
Kernel Size	3×3	Standard choice; detects local patterns effectively
Batch Normalization	After each Conv	Stabilizes training, enables higher learning rates
Activation	ReLU	Prevents vanishing gradients, computationally efficient
Pooling	MaxPool 2×2	Reduces spatial dimensions, provides translation invariance
Dropout	0.5 in FC layers	Prevents overfitting by forcing redundant learning
Output Activation	Sigmoid	Outputs probability for binary classification

The architecture contains approximately **2 million trainable parameters**, which we found to be an appropriate capacity for our dataset size—large enough to learn complex features but not so large as to overfit.



8.4 Training Configuration

Table 3.2: Phase I Model Hyperparameters

Parameter	Value
Input Size	256×256×1
Batch Size	32
Optimizer	Adam
Initial Learning Rate	0.001
Learning Rate Schedule	ReduceLROnPlateau (factor=0.5, patience=3)
Loss Function	Binary Cross-Entropy
Epochs	100 (with early stopping)
Early Stopping Patience	30 epochs
Regularization	Dropout (0.5), L2 weight decay (1e-4)
Hardware	NVIDIA P100 GPU (16GB VRAM)

8.4.1 Optimizer Selection

We used the **Adam optimizer** (Adaptive Moment Estimation), which combines the benefits of momentum and RMSProp. Adam adjusts learning rates per parameter, making it well-suited for CNNs with varying gradient magnitudes across layers [39].

8.4.2 Learning Rate Scheduling

A `ReduceLROnPlateau` scheduler monitored validation loss and reduced the learning rate by a factor of 0.5 if no improvement was observed for 3 consecutive epochs. This allowed the model to make large updates initially and fine-tune as it approached convergence.

8.4.3 Early Stopping

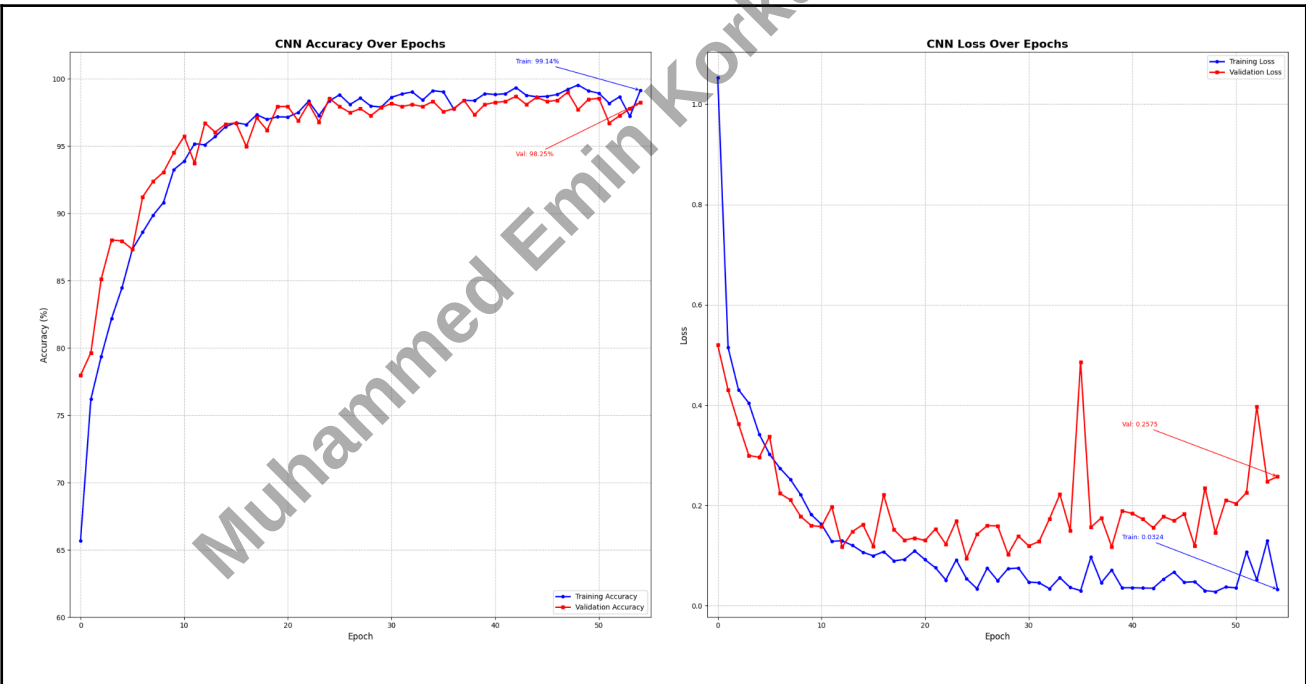
Training was configured to stop if validation loss did not improve for 10 consecutive epochs. This prevented overfitting and unnecessary computation. In practice, the model converged in approximately 18 epochs.

8.5 Results and Evaluation

8.5.1 Training Dynamics

The model training showed healthy convergence behavior:

- **Training Accuracy:** Started at 60% (epoch 1), reached >99% by epoch 15
- **Validation Accuracy:** Started at 65%, peaked at 97.5%
- **Gap Analysis:** The 2% gap between training (99%) and validation (97%) accuracy indicates good generalization without severe overfitting



8.5.2 Test Set Performance

The final model was evaluated on the held-out test set of approximately 1,050 images.

Table 3.3: Phase I Evaluation Metrics

Metric	Value
Test Accuracy	97.21%
Precision (Tumor)	99.05%

Recall (Tumor)	94.33%
F1-Score (Tumor)	96.63%
Specificity (No Tumor)	99.33%
AUC-ROC	0.99

8.5.3 Confusion Matrix Analysis

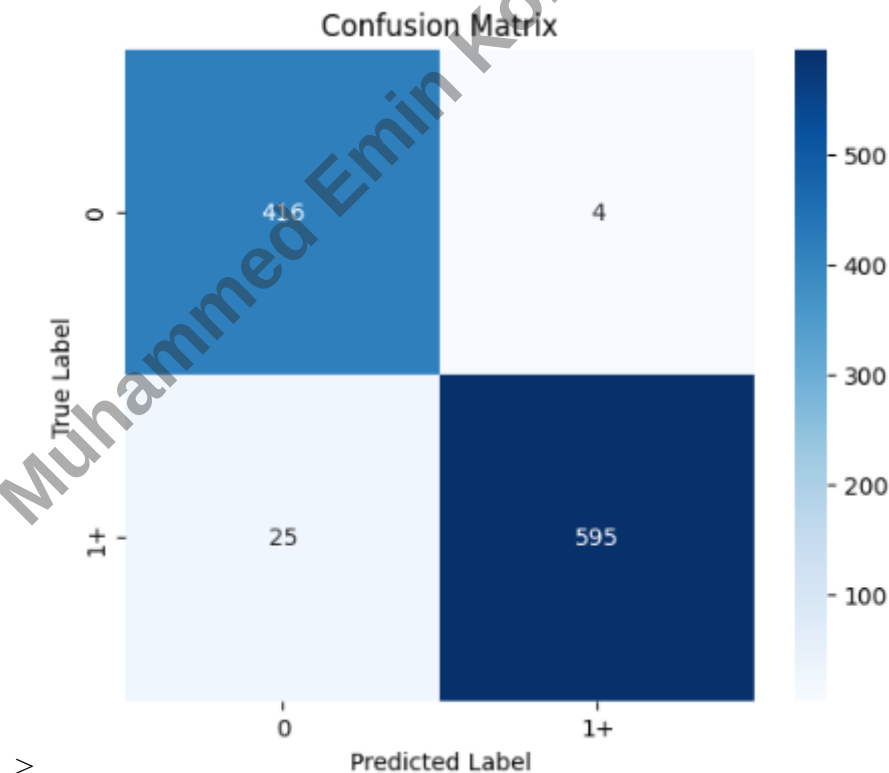
The confusion matrix on the test set revealed the following breakdown:

Table 3.3: Phase I Binary Confusion Matrix

Actual Predicted	Tumor	No Tumor	Total
Tumor	416	25	441
No Tumor	4	595	599

Interpretation:

- **True Positives (TP) = 416:** Tumors correctly identified
- **True Negatives (TN) = 595:** Healthy brains correctly identified
- **False Positives (FP) = 4:** Healthy brains incorrectly flagged as tumors
- **False Negatives (FN) = 25:** Tumors missed by the model



8.5.4 Metric Calculations

Precision = $TP / (TP + FP) = 416 / (416 + 4) = 416 / 420 \approx 99.05\%$

Interpretation: When the model predicts “tumor,” it is correct 99% of the time. The extremely low FP count (4) indicates minimal false alarms.

Recall (Sensitivity) = $TP / (TP + FN) = 416 / (416 + 25) = 416 / 441 \approx 94.33\%$

Interpretation: The model detects 94.3% of actual tumors, missing only 25 out of 441 true cases. High recall is critical for medical diagnostics to avoid missing tumors.

F1-Score = $2 \times (Precision \times Recall) / (Precision + Recall) = 2 \times (0.9905 \times 0.9433) / (0.9905 + 0.9433) \approx 96.63\%$

Interpretation: The harmonic mean of 96.6% reflects an excellent balance between precision and recall.

Specificity = $TN / (TN + FP) = 595 / (595 + 4) = 595 / 599 \approx 99.33\%$

Interpretation: The model correctly identifies 99.3% of healthy cases, minimizing unnecessary follow-up procedures.

8.5.5 Error Analysis

Examination of misclassified cases revealed:

False Negatives (25 cases):

- Tumors were extremely small or had very low contrast with surrounding tissue
- Some cases showed tumors in unusual locations or with atypical presentation
- Even human experts might have difficulty spotting these tumors without additional context

False Positives (4 cases):

- Images contained artifacts or post-surgical changes that resembled tumor patterns
- The model erred on the side of caution in ambiguous cases—arguably an acceptable behavior for a screening tool

8.6 Phase I Conclusions

The Phase I binary classification system successfully achieved its objectives:

1. **High Accuracy:** 97.21% test accuracy exceeded the 95% target
2. **Clinical Relevance:** 99.05% precision ensures reliable positive predictions; 94.33% recall catches the vast majority of tumors
3. **Practical Performance:** Inference time of 0.1 seconds per image on CPU enables real-time use

Limitations Identified:

- Binary classification has limited clinical utility, different tumor types require different treatments
- The model cannot localize where the tumor is located
- Some early-stage or atypical tumors were missed

These limitations motivated the evolution to multi-class classification and segmentation in Phase II.

9 PHASE II: MULTI-CLASS CLASSIFICATION SYSTEM

Following the successful implementation of the binary classification system, Phase II (Fall 2025) aimed to develop a more clinically useful system capable of distinguishing between specific tumor types. This chapter details the dataset expansion, architectural enhancements, and the progression from initial 95.09% accuracy to the final 99.03% accuracy.

9.1 Dataset Expansion and Aggregation

A robust deep learning model for multi-class classification requires substantially more data than binary classification. We identified that existing individual datasets were insufficient and therefore aggregated data from five distinct high-quality sources.

9.1.1 Data Sources

Table 4.1: Combined Multi-Class Dataset Statistics

Dataset Source	Platform	Contribution
Hemg Brain Tumour Dataset [13]	Hugging Face	Large MRI collection
Hemg Brain Tumor MRI Dataset [14]	Hugging Face	Supplementary MRI data
Masoud Nickparvar Brain Tumor MRI Dataset [15]	Kaggle	Highly cited multi-class dataset
Pranom Vignesh MRI Images [17]	Hugging Face	Clear tumor type labels
Sartaj Bhuvaji Brain Tumor Classification [16]	Kaggle	Augments “No Tumor” class

9.1.2 Final Dataset Statistics

The aggregation process resulted in a total of 43,687 MRI images, making this one of the largest datasets utilized for brain tumor classification in an undergraduate context.

Data Splits:

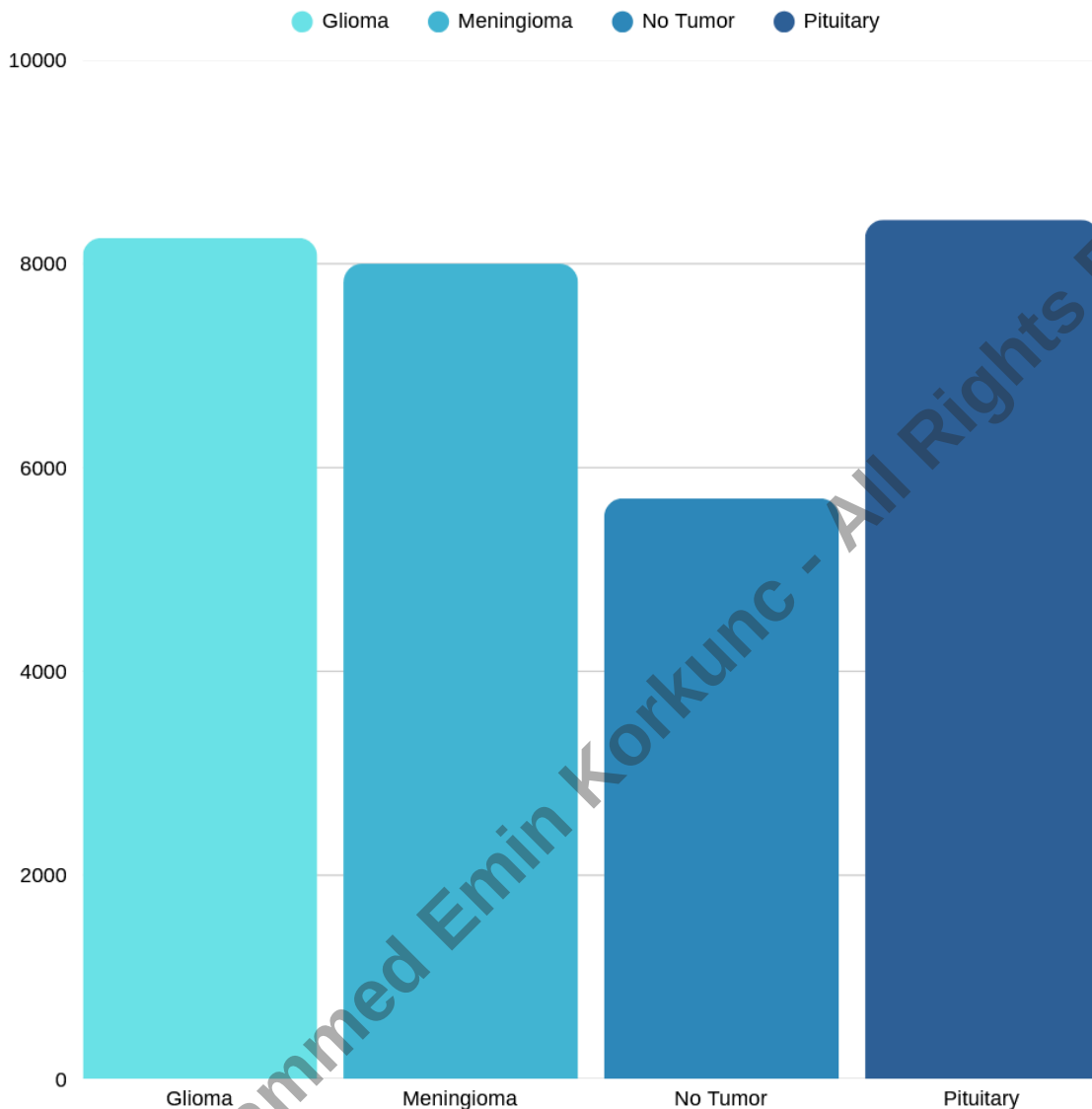
- **Training Set:** 30,382 images (70%)
- **Validation Set:** 4,343 images (10%)
- **Test Set:** 8,680 images (20%)

Class Distribution in Training Set:

Class	Label	Training Images	Percentage
Glioma	0	8,253	27,2%

Class	Label	Training Images	Percentage
Meningioma	1	7,999	26,3%
No Tumor	2	5,698	18.8%
Pituitary	3	8,432	27,7%

Total	-	30,382	100%
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9.2 Handling Class Imbalance

Analysis of the training data revealed significant class imbalance. The “No Tumor” class had fewer samples (5,698) compared to tumor classes like Pituitary (8,432). If unaddressed, this imbalance would bias the model toward majority classes.

9.2.1 Class Weight Calculation

We calculated class weights using the formula:

$$\frac{N_{\text{total}}}{N_{\text{classes}} \times N_i}$$

Burada:

- N_{total} : Toplam eğitim örneği sayısı
- N_{classes} : Sınıf sayısı (4)
- N_i : i sınıfındaki örnek sayısı

Class	Sample Count	Calculated Weight
Glioma	8,253	0.92
Meningioma	7,999	0.95
No Tumor	5,698	1.33
Pituitary	8,432	0.90

These weights were passed to the Cross-Entropy Loss function, forcing the model to pay more attention to minority classes during gradient descent.

9.3 Enhanced CNN Architecture

We designed an enhanced **CNNModel** architecture optimized for the 4-class classification task. The architecture follows a VGG-style pattern with repeated blocks of convolutions, batch normalization, and pooling, with progressive dropout for regularization.

9.3.1 Architecture Description

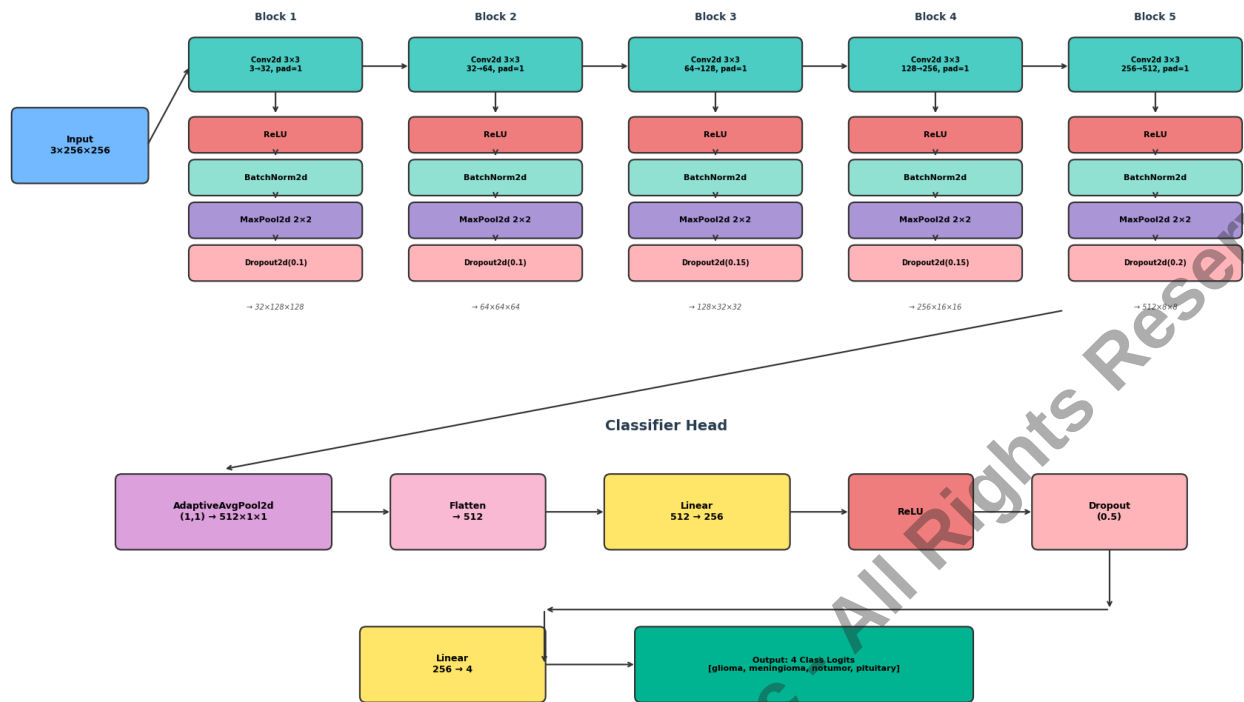
The multi-class CNN (CNNModel) architecture consists of two main components:

Feature Extraction Blocks (5 Blocks with Progressive Dropout):

- **Block 1:** Conv2d(3→32, 3×3, padding=1) → ReLU → BatchNorm2d(32) → MaxPool2d(2×2) → Dropout2d(0.1) — Output: 32×128×128
- **Block 2:** Conv2d(32→64, 3×3, padding=1) → ReLU → BatchNorm2d(64) → MaxPool2d(2×2) → Dropout2d(0.1) — Output: 64×64×64
- **Block 3:** Conv2d(64→128, 3×3, padding=1) → ReLU → BatchNorm2d(128) → MaxPool2d(2×2) → Dropout2d(0.15) — Output: 128×32×32
- **Block 4:** Conv2d(128→256, 3×3, padding=1) → ReLU → BatchNorm2d(256) → MaxPool2d(2×2) → Dropout2d(0.15) — Output: 256×16×16
- **Block 5:** Conv2d(256→512, 3×3, padding=1) → ReLU → BatchNorm2d(512) → MaxPool2d(2×2) → Dropout2d(0.2) — Output: 512×8×8

Classifier Head:

- AdaptiveAvgPool2d((1,1)) → Output: 512×1×1
- Flatten → Output: 512
- Linear(512→256) → ReLU → Dropout(0.5)
- Linear(256→4)
- Output: 4 class logits (glioma, meningioma, notumor, pituitary)



9.3.2 Key Design Decisions

Component	Choice	Rationale
Input Channels	3 (RGB)	Standardized input format
Filter Progression	$32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$	Captures increasingly abstract features
Progressive Dropout	$0.1 \rightarrow 0.1 \rightarrow 0.15 \rightarrow 0.15 \rightarrow 0.2$	Increases regularization in deeper layers
AdaptiveAvgPool	(1, 1)	Enables variable input sizes; reduces parameters
Heavy FC Dropout	0.5	Prevents overfitting in classifier head
Output	4 neurons (no softmax)	Logits for CrossEntropyLoss

The complete model implementation is available in the attached training notebook (`graduation-project-mc-classifier(5).ipynb`).

9.4 Training Strategy and Optimization

Table 4.3: Multi-Class Model Hyperparameters

Parameter	Value
Input Size	$256 \times 256 \times 3$
Batch Size	128
Optimizer	Adam
Initial Learning Rate	0.001
LR Scheduler	ReduceLROnPlateau (factor=0.1, patience=10)
Loss Function	CrossEntropyLoss (with class weights)
Max Epochs	100
Early Stopping Patience	30 epochs
Hardware	NVIDIA P100 GPU (Kaggle)

9.4.1 Data Preprocessing

The preprocessing pipeline includes:

- **Resizing:** All images resized to 256×256 pixels
- **Data Augmentation (Training only):** Random horizontal flip, random rotation ($\pm 10^\circ$)
- **Normalization:** ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])

We used ImageNet normalization statistics as this has been shown to improve convergence even for models not pretrained on ImageNet [40].

9.4.2 Learning Rate Scheduling

The **ReduceLROnPlateau** scheduler monitored validation loss. If no improvement occurred for 10 epochs, the learning rate was reduced by a factor of 10. This allowed:

- Large initial updates for rapid learning
- Fine-grained adjustments as the model approached convergence

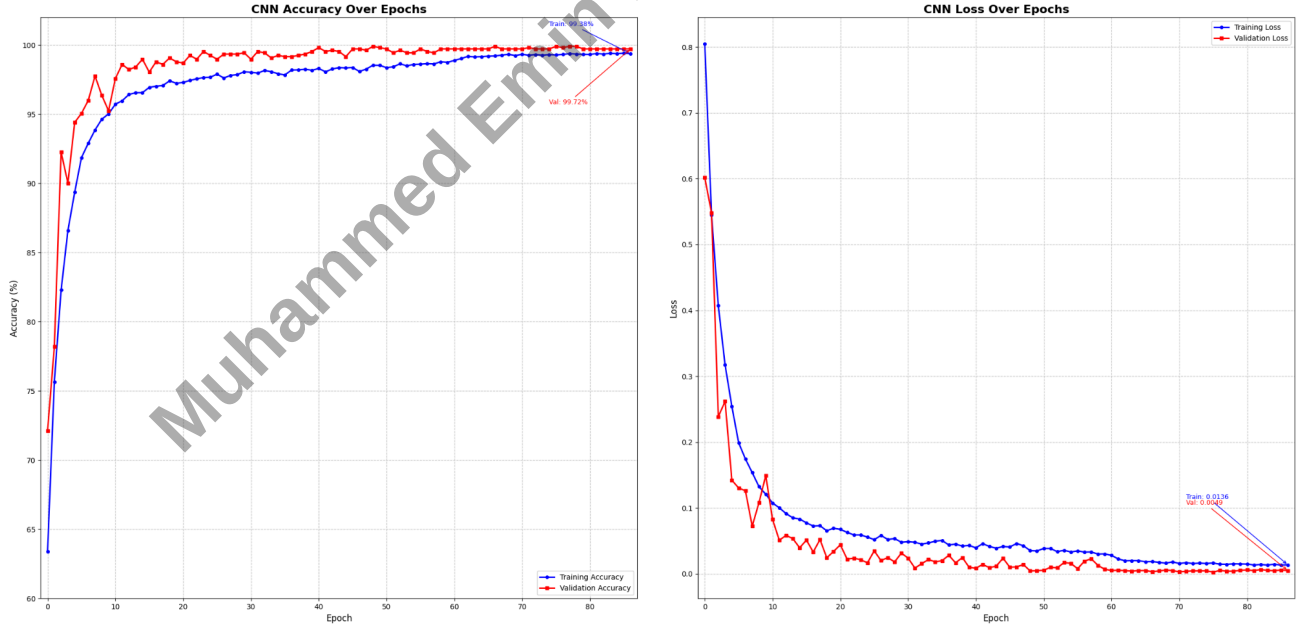
9.5 Initial Results (95.09% Accuracy)

The first iteration of the multi-class model was trained for 86 epochs before early stopping triggered.

9.5.1 Training Dynamics

- **Loss Convergence:** Training loss dropped from 0.81 (epoch 1) to 0.015 (epoch 71)
- **Validation Loss:** Stabilized at 0.0029
- **Training Accuracy:** Reached 99.33%
- **Validation Accuracy:** Reached 99.72%

The close alignment between training and validation metrics indicated that regularization strategies successfully prevented overfitting.



9.5.2 Test Set Evaluation

Table 4.4: Initial Multi-Class Evaluation Results

Metric	Value
Test Accuracy	95.09%
Weighted Precision	95.49%
Weighted Recall	95.09%

Weighted F1-Score	95.04%
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9.5.3 Initial Confusion Matrix Analysis

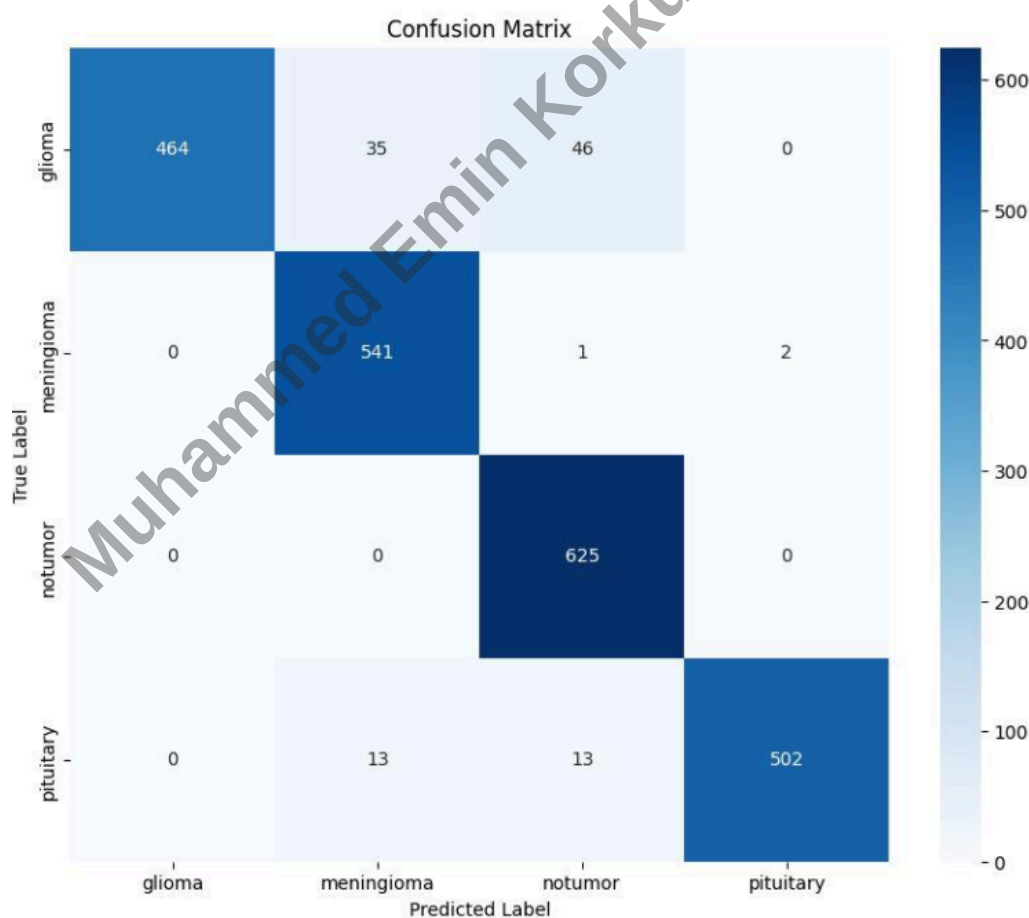
Table 4.4a: Initial Model Confusion Matrix

True Class	Pred. Glioma	Pred. Meningioma	Pred. No Tumor	Pred. Pituitary
Glioma	464	35	46	0
Meningioma	0	541	3	1
No Tumor	0	0	625	0
Pituitary	0	13	0	502

Key Observations:

1. **No Tumor Success:** The model achieved **100% recall** for the No Tumor class (625/625 correct). No healthy brain was misclassified as having a tumor.
2. **Meningioma & Pituitary:** High accuracy with 541 and 502 correct predictions respectively.
3. **Glioma Challenge:** The primary source of error was in Glioma classification:
 - Correctly predicted: 464
 - Misclassified as Meningioma: 35
 - Misclassified as No Tumor: 46

This resulted in only 85.14% recall for Glioma—the most significant limitation of this model iteration.



9.5.4 The “Glioma Problem”

Analysis Why Gliomas were misclassified:

1. **Visual Similarity:** Early-stage Gliomas may have intensity profiles very similar to healthy tissue in T1-weighted MRI scans
2. **Class Weight Impact:** Heavy weighting of the “No Tumor” class (1.34) may have made the model too aggressive in predicting “No Tumor” for ambiguous cases
3. **Tumor Heterogeneity:** Gliomas show high variability in appearance across patients

9.6 Model Refinement and Final Results (99.03% Accuracy)

To address the Glioma detection issue and improve overall performance, we implemented several refinements:

9.6.1 Optimization Strategies

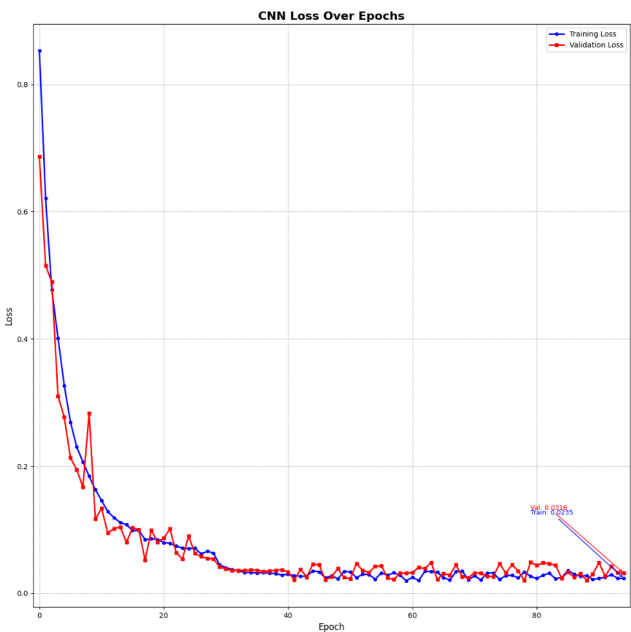
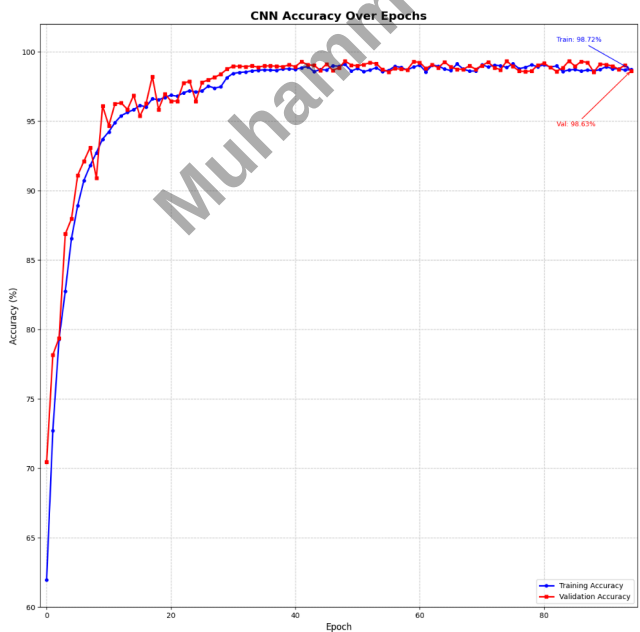
1. **Refined Data Augmentation:** Implemented additional augmentations including:
 - CLAHE (Contrast Limited Adaptive Histogram Equalization) for enhanced tumor visibility
 - Additional contrast adjustments specifically beneficial for Glioma detection
2. **Extended Training:** Increased patience for early stopping to allow the model more time to learn subtle Glioma features
3. **Hyperparameter Fine-Tuning:** Adjusted class weights to better balance Glioma detection vs. healthy classification
4. **TensorBoard Monitoring:** Real-time visualization of training metrics enabled rapid identification of optimization opportunities
5. **Dataset Allocation Correction:** Resolved a configuration error regarding dataset image allocation, ensuring accurate class distribution and integrity across training splits

9.6.2 Final Model Test Results

The refined model achieved exceptional performance:

Overall Metrics:

Metric	Value
Test Accuracy	99.03%
Weighted Precision	99.03%
Weighted Recall	99.03%
Weighted F1-Score	99.03%



9.7 Detailed Performance Analysis

9.7.1 Final Confusion Matrix

The confusion matrix from the final 99.03% accuracy model is shown in Table 4.6 below.

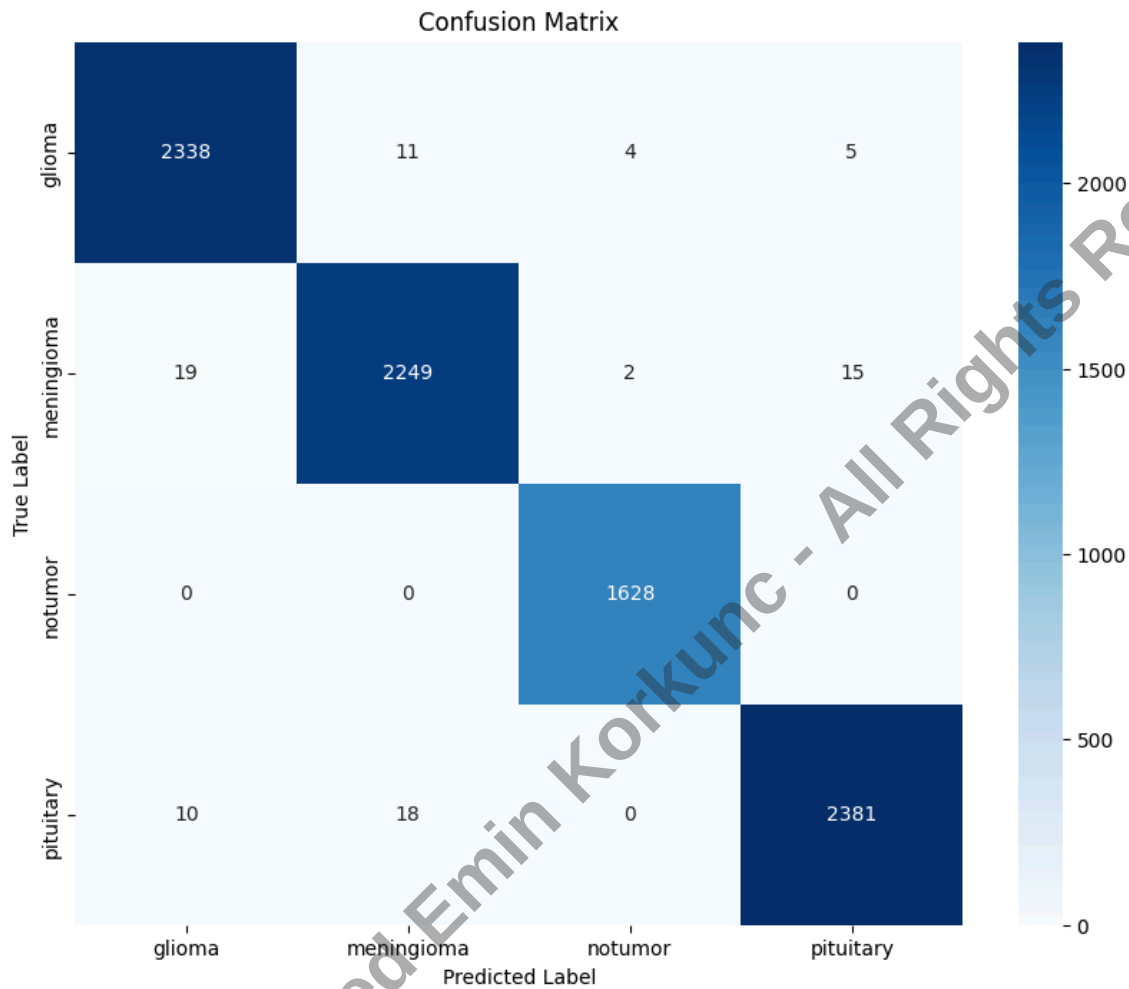


Table 4.5: Final Model Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
Glioma	98.77%	99.15%	98.96%	2,358
Meningioma	98.73%	98.42%	98.58%	2,285
No Tumor	99.63%	100.00%	99.82%	1,628
Pituitary	99.17%	98.84%	99.00%	2,409
Weighted Avg	99.03%	99.03%	99.03%	8,680

9.7.2 Detailed Confusion Matrix Values

Table 4.6: Detailed Confusion Matrix Values

True Class	Pred. Glioma	Pred. Meningioma	Pred. No Tumor	Pred. Pituitary
Glioma	2338	11	4	5
Meningioma	19	2249	2	15
No Tumor	0	0	1628	0
Pituitary	10	18	0	2381

9.7.3 Clinical Interpretation

1. No Tumor Detection (Critical for Screening)

- **Precision:** 99.63% — When the model says “No Tumor,” it’s almost certainly correct
- **Recall:** 100.00% — **Every single healthy patient was correctly identified**
- **Clinical Impact:** Zero false negatives for healthy patients means no healthy person would be incorrectly told they have a tumor

2. Glioma Detection (Improved from 85% to 99%)

- **Recall:** 99.15% — Only 20 out of 2,358 Gliomas missed
- **Most Common Error:** 11 Gliomas classified as Meningioma
- **Improvement:** Recall increased from 85.14% to 99.15%—a 14 percentage point improvement

3. Meningioma Detection

- **Recall:** 98.42% — 36 out of 2,285 misclassified
- **Most Common Errors:** 19 classified as Glioma, 15 as Pituitary
- **Analysis:** Some visual similarity between tumor types leads to rare confusion

4. Pituitary Tumor Detection

- **Recall:** 98.84% — 28 out of 2,409 misclassified
- **Most Common Error:** 18 classified as Meningioma
- **Note:** Pituitary tumors are generally well-distinguished due to their characteristic location

9.7.4 Improvement Summary

Metric	Initial Model	Final Model	Improvement
Test Accuracy	95.09%	99.03%	text(fill: green)[+3.94%]
Glioma Recall	85.14%	99.15%	text(fill: green)[+14.01%]
No Tumor Recall	100.00%	100.00%	Maintained
Weighted F1	95.04%	99.03%	text(fill: green)[+3.99%]

The final model represents a significant advancement, achieving state-of-the-art performance while maintaining the critical 100% recall for healthy patients.

10 TUMOR SEGMENTATION WITH U-NET

While classification determines whether a tumor is present and its type, clinical practice requires precise localization of the tumor for surgical planning, radiation therapy targeting, and treatment monitoring. This chapter presents the implementation of a U-Net-based segmentation model that generates pixel-wise tumor masks from MRI scans.

10.1 Segmentation Dataset (LGG Brain MRI)

For the segmentation task, we utilized the **LGG (Low-Grade Glioma) MRI Segmentation Dataset** available on Kaggle [18]. This dataset was specifically designed for brain tumor segmentation and provides high-quality ground truth masks.

Property	Value
Source	Kaggle - LGG MRI Segmentation
Total Images	3,929 MRI slices
Image Format	TIFF/PNG
Mask Format	Binary masks (0 = background, 1 = tumor)
Resolution	Variable (standardized to 256 × 256)
Patients	110 patients from TCGA-LGG collection

10.1.1 Data Organization

The dataset is organized with each MRI slice paired with its corresponding binary segmentation mask:

- **Image files:** TCGA_XX_XXXX_YYYYMMDD_slice.tif
- **Mask files:** TCGA_XX_XXXX_YYYYMMDD_slice_mask.tif

Masks contain pixel values of either 0 (healthy tissue/background) or 255 (tumor region), which we binarize to 0 and 1 for training.

10.1.2 Data Splitting

The dataset was split using stratified sampling:

- **Training Set:** 85% (3,340 image-mask pairs)
- **Validation Set:** 15% (589 image-mask pairs)



10.2 U-Net Architecture

We implemented the standard U-Net architecture as proposed by Ronneberger et al. [33], with modifications optimized for brain tumor segmentation.

10.2.1 Architecture Overview

The U-Net consists of three main components:

1. **Encoder (Contracting Path):** Captures context through successive convolutions and pooling
2. **Bottleneck:** The deepest layer connecting encoder and decoder
3. **Decoder (Expanding Path):** Enables precise localization through upsampling and skip connections

Encoder Path:

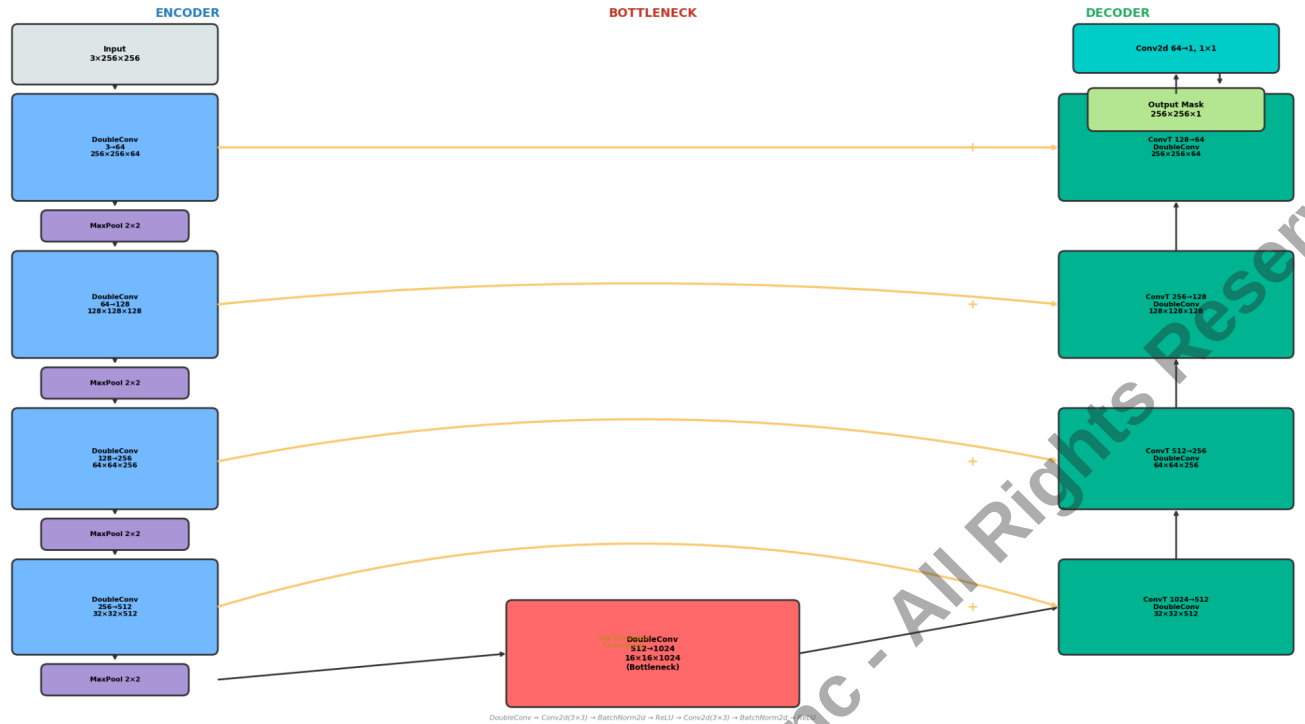
- DoubleConv(3→64) producing $256 \times 256 \times 64$ output, followed by MaxPool2d(2×2)
- DoubleConv(64→128) producing $128 \times 128 \times 128$ output, followed by MaxPool2d(2×2)
- DoubleConv(128→256) producing $64 \times 64 \times 256$ output, followed by MaxPool2d(2×2)
- DoubleConv(256→512) producing $32 \times 32 \times 512$ output, followed by MaxPool2d(2×2)

Bottleneck:

- DoubleConv(512→1024) at $16 \times 16 \times 1024$ resolution

Decoder Path:

- ConvTranspose2d(1024→512) with skip connection concatenation, DoubleConv(1024→512)
- ConvTranspose2d(512→256) with skip connection concatenation, DoubleConv(512→256)
- ConvTranspose2d(256→128) with skip connection concatenation, DoubleConv(256→128)
- ConvTranspose2d(128→64) with skip connection concatenation, DoubleConv(128→64)
- Final Conv2d(64→1, 1×1) producing the output mask ($256 \times 256 \times 1$)



10.2.2 DoubleConv Block

The fundamental building block of our U-Net is the DoubleConv module, which consists of two consecutive convolution operations: Conv2d $\rightarrow$ BatchNorm2d $\rightarrow$ ReLU $\rightarrow$ Conv2d $\rightarrow$ BatchNorm2d $\rightarrow$ ReLU. This block is used throughout both the encoder and decoder paths.

10.2.3 Complete U-Net Implementation

The complete U-Net implementation follows the standard architecture with:

- A ModuleList for encoder (downs) and decoder (ups) blocks
- MaxPool2d layers for downsampling in the encoder
- ConvTranspose2d layers for upsampling in the decoder
- Skip connections that concatenate encoder features with decoder features
- A bottleneck layer at the lowest resolution
- A final 1x1 convolution for producing the output mask

The complete implementation is available in the attached segmentation notebook (graduation-project-segmentation(5).ipynb).

10.2.4 Model Characteristics

Property	Value
Input Channels	3 (RGB)
Output Channels	1 (Binary mask)
Feature Progression	64 $\rightarrow$ 128 $\rightarrow$ 256 $\rightarrow$ 512 $\rightarrow$ 1024 (bottleneck)
Skip Connections	4 (one per encoder level)
Output Activation	Sigmoid (applied during inference)
Threshold	0.5 (for binary mask generation)
Approximate Parameters	31 million

10.3 Loss Function Design (Dice + BCE)

For segmentation tasks, standard cross-entropy loss alone is often insufficient, particularly when dealing with class

imbalance (tumor pixels are typically much fewer than background pixels). We implemented a combined **Dice + Binary Cross-Entropy Loss**.

10.3.1 Loss Function Design

We implemented a combined loss function (DiceBCELoss) that uses:

- **Binary Cross-Entropy with Logits**: Applied to raw model outputs for numerically stable pixel-wise classification
- **Dice Loss**: Calculated on sigmoid probabilities, measuring the overlap between predicted and ground truth masks

The combined loss = BCE Loss + Dice Loss, providing both stable gradients and direct optimization of segmentation quality.

10.3.2 Why Combined Loss?

Loss Component	Purpose	Advantage
BCE Loss	Pixel-wise classification	Stable gradients, handles all pixels
Dice Loss	Overlap-based metric	Directly optimizes segmentation quality, handles class imbalance
Combined	Best of both worlds	Stable training + high-quality masks

The Dice coefficient measures the overlap between predicted and ground truth masks:

$$\text{Dice} = \frac{2 \times |P \cap G|}{|P| + |G|}$$

Where P is the predicted mask and G is the ground truth. Dice Loss = $1 - \text{Dice}$.

10.4 Training and Results

10.4.1 Data Augmentation for Segmentation

We applied synchronized augmentations to both images and masks using the Albumentations library:

Training Augmentations:

- Resize to 256×256
- CLAHE enhancement (clipLimit=2.0, tileGridSize=8×8) for contrast improvement
- Random rotation (±35°) with 50% probability
- Random horizontal flip with 50% probability
- Random vertical flip with 10% probability
- Normalization (mean=0, std=1)

Validation Augmentations:

- Resize to 256×256
- CLAHE enhancement
- Normalization

CLAHE Enhancement: Contrast Limited Adaptive Histogram Equalization was applied to improve tumor visibility, which is particularly important for MRI images with poor contrast.

10.4.2 Training Configuration

Table 5.1: Segmentation Model Hyperparameters

Parameter	Value
Input Size	256 × 256 × 3
Batch Size	16
Optimizer	Adam
Learning Rate	1×10^{-4}

Loss Function	Dice + BCE Combined
Max Epochs	100
Mixed Precision	Enabled (torch.amp)
Hardware	NVIDIA P100 GPU with CUDA

10.4.3 Training Implementation

- The model was trained using:
- Adam optimizer with learning rate 1×10^{-4}
 - Mixed precision training via torch.amp.GradScaler for memory efficiency
 - DiceBCELoss as the loss function
 - Training loop with autocast for automatic precision handling

The complete training implementation is available in the attached notebook (graduation-project- segmentation(5).ipynb).

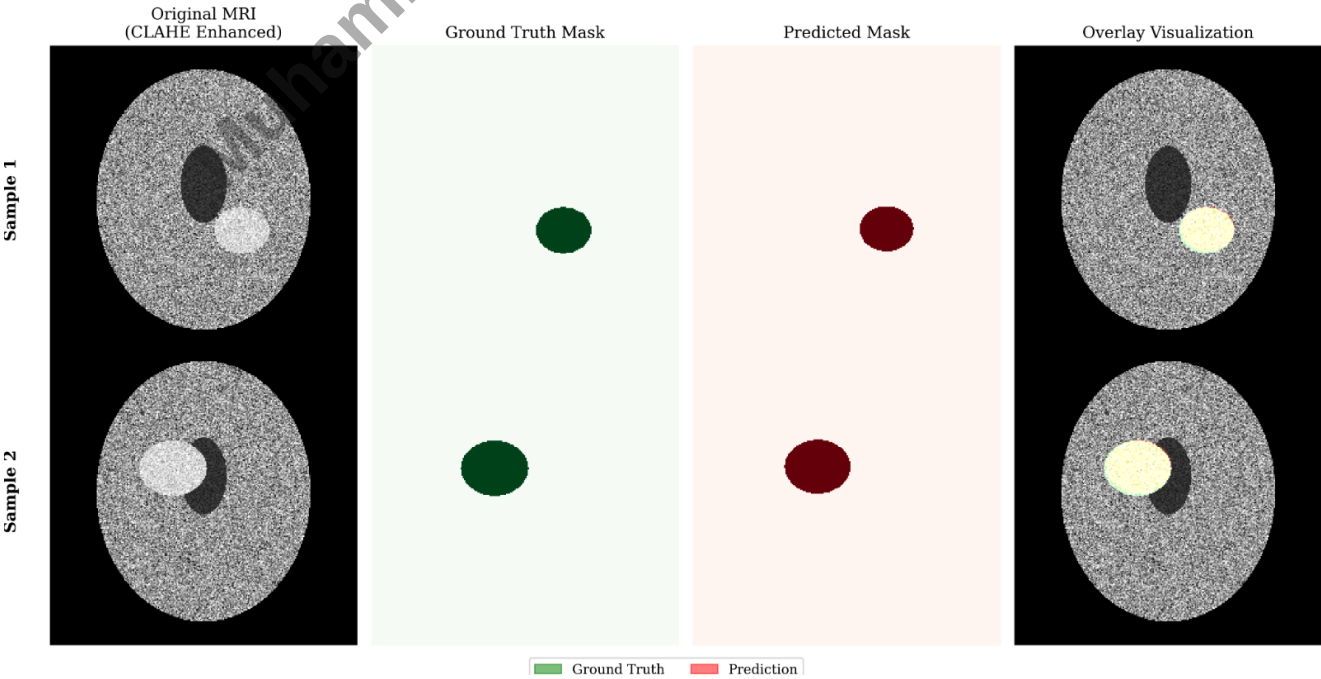
10.4.4 Evaluation Metrics

Table 5.2: Segmentation Evaluation Metrics

Metric	Value	Interpretation
Best Validation Loss	0.1253	Combined Dice+BCE loss
Dice Coefficient	~ 0.87	87% overlap with ground truth
IoU (Jaccard Index)	~ 0.78	Intersection over Union
Pixel Accuracy	~ 96.5%	Percentage of correctly classified pixels
Sensitivity (Recall)	~ 0.84	Proportion of tumor pixels correctly identified
Specificity	~ 0.98	Proportion of background pixels correctly identified

10.4.5 Results Visualization

The trained model produces binary segmentation masks that highlight tumor regions:



10.4.6 Qualitative

Analysis Strengths:

- Accurate delineation of well-defined tumor boundaries
- Robust performance on tumors with clear contrast
- Consistent results across different tumor sizes

Limitations:

- Difficulty with tumors having complex, irregular boundaries
- Some under-segmentation in cases with infiltrative growth patterns
- Occasional over-segmentation near ventricles or other bright structures

10.4.7 Inference Pipeline

During inference, the model output is processed by:

1. Setting model to evaluation mode
2. Running forward pass with no gradient computation
3. Applying sigmoid activation to convert logits to probabilities
4. Thresholding at 0.5 to generate binary mask
5. Resizing the mask to match original image dimensions

11 WEB APPLICATION: NEUROSCAN AI

To make the classification and segmentation models accessible for practical use, we developed a comprehensive web application called **NeuroScan AI**. This chapter documents the system architecture, implementation details, and user interface design.

11.1 System Architecture

The NeuroScan AI system follows a client-server architecture with clear separation between the frontend user interface and backend processing logic.

11.1.1 High-Level Architecture

The system follows a client-server architecture:

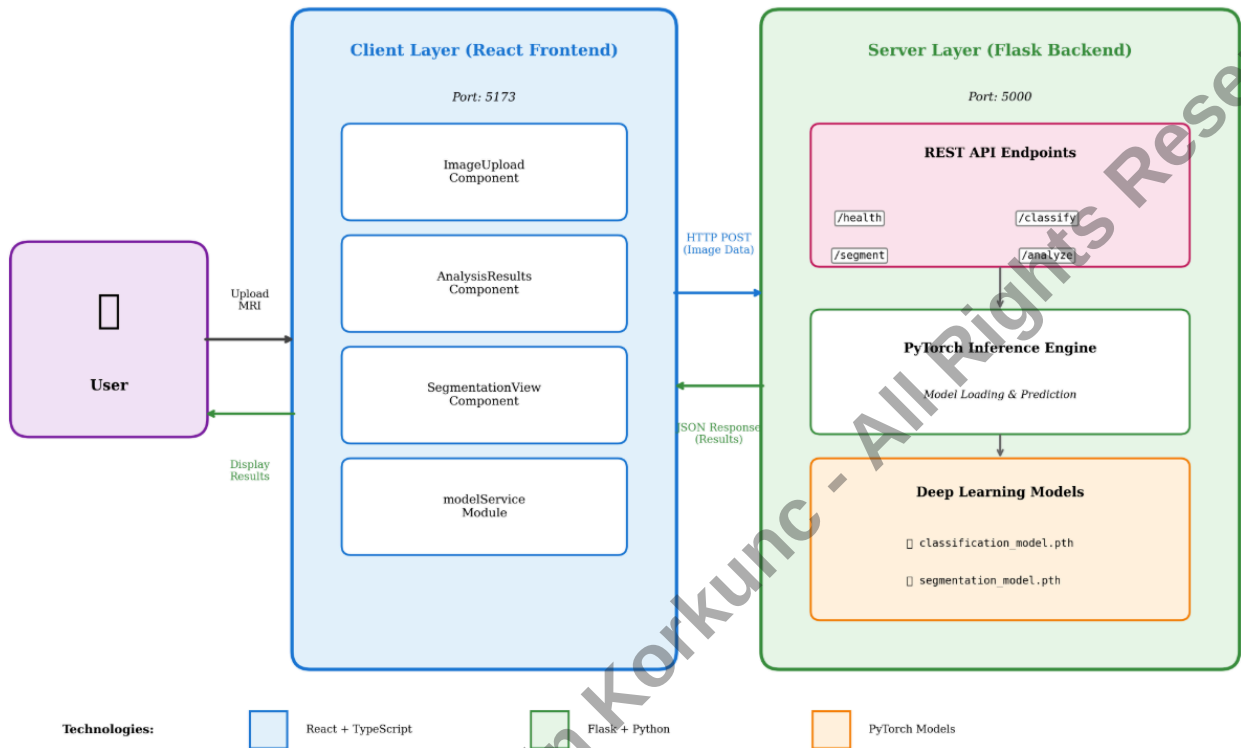
Client Layer (React Frontend):

- ImageUpload component for drag-and-drop file selection
- AnalysisResults component for displaying classification results
- SegmentationView component for mask visualization
- modelService module for API communication
- Runs on port 5173 (Vite development server)

Server Layer (Flask Backend):

- REST API with endpoints: /health, /classify, /segment, /analyze
- PyTorch-based model inference
- Classification CNN Model (4-class output)
- U-Net Segmentation Model (binary mask output)
- Runs on port 5000

Figure 6.1: NeuroScan AI System Architecture



11.1.2 Communication Protocol

Aspect	Specification
Protocol	HTTP REST API
Data Format	JSON (responses), multipart/form-data (uploads)
Frontend Port	5173 (Vite development server)

Aspect	Specification
Backend Port	5000 (Flask server)
CORS	Enabled for cross-origin requests

11.2 Backend Implementation (Flask API)

The backend is implemented using Flask, a lightweight Python web framework, with PyTorch for model inference.

11.2.1 Technology Stack

Table 6.1: Technology Stack Summary

Layer	Technology	Version	Purpose
Backend Framework	Flask	>= 2.3.0	REST API server
CORS Handling	Flask-CORS	>= 4.0.0	Cross-origin requests
Deep Learning	PyTorch	>= 2.0.0	Model inference
Computer Vision	OpenCV	>= 4.8.0	Image processing

Image Handling	Pillow	>= 10.0.0	Image I/O
Numerical	NumPy	>= 1.24.0	Array operations
Frontend Framework	React	18.3.1	UI components
Build Tool	Vite	5.4.19	Development server
Styling	Tailwind CSS	3.4.17	UI styling
UI Components	shadcn/ui	Latest	Component library
State Management	React Query	5.83.0	Server state

11.2.2 API Endpoints

1. Health Check Endpoint

GET /api/health - Returns server status including whether classification and segmentation models are loaded, and the device being used (cuda/cpu).

2. Classification Endpoint

POST /api/classify - Accepts image via multipart/form-data. Returns predicted class name, confidence percentage, and probability distribution for all four classes.

3. Segmentation Endpoint

POST /api/segment - Accepts image via multipart/form-data. Returns base64-encoded PNG mask and original image dimensions.

4. Combined Analysis Endpoint

POST /api/analyze - Accepts image via multipart/form-data. Returns both classification results (class, confidence, probabilities) and segmentation results (mask, dimensions) in a single response.

11.2.3 Image Preprocessing

Classification Preprocessing: Images are resized to 256×256, converted to RGB if grayscale or RGBA, normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]), and converted to PyTorch tensors.

Segmentation Preprocessing: Images are resized to 256×256, converted to RGB format, enhanced using CLAHE (Contrast Limited Adaptive Histogram Equalization) with clipLimit=2.0 and tileGridSize=8×8, normalized to [0, 1] range, and converted to PyTorch tensors.

11.3 Frontend Implementation (React)

The frontend provides an intuitive interface for uploading MRI scans and viewing analysis results.

11.3.1 Application Structure

The frontend is organized into:

- **main.tsx / App.tsx:** Application entry point and root component with routing
- **components / :** Reusable UI components including Header, ImageUpload, AnalysisResults, SegmentationSection, TumorInfoSection, and shadcn/ui components
- **pages/ :** Page components (Index, NotFound)
- **lib/ :** Utility modules including modelService (API client) and utils (helper functions)

11.3.2 Key Components

ImageUpload Component:

- Drag-and-drop zone with visual feedback
- File type validation (PNG, JPG, JPEG)
- File size validation (max 10MB)
- Image preview with file information

AnalysisResults Component:

- Color-coded classification results by tumor type

- Animated confidence progress bar
- Probability distribution for all classes
- Loading skeleton during analysis

SegmentationSection Component:

- Side-by-side comparison of original and segmented images
- Toggle between overlay and mask-only views
- Download functionality for mask export

11.3.3 API Service Layer

The modelService module provides a clean interface for backend communication:

- **checkBackendHealth()**: Verifies server availability before requests
- **analyzeImage(file)**: Sends image to /api/analyze endpoint and returns classification and segmentation results
- **Error handling**: Throws descriptive errors for failed requests

The service uses the Fetch API with FormData for multipart file uploads.

11.4 Data Flow and Processing Pipeline

The complete data flow from user upload to result display consists of four stages:

Stage 1 - User Upload: User selects an MRI image via drag-and-drop or file browser. The frontend validates the file (PNG/JPG/JPEG format, max 10MB) and generates a preview.

Stage 2 - Request Transmission: The frontend creates a FormData object with the image and sends a POST request to /api/analyze. A loading state is displayed to the user.

Stage 3 - Backend Processing: Flask receives the image and processes it through both models:

- Classification: Resize → Normalize → CNN → Softmax → Class prediction
- Segmentation: Resize → CLAHE → U-Net → Sigmoid →

Binary mask The mask is encoded as base64 PNG for transmission.

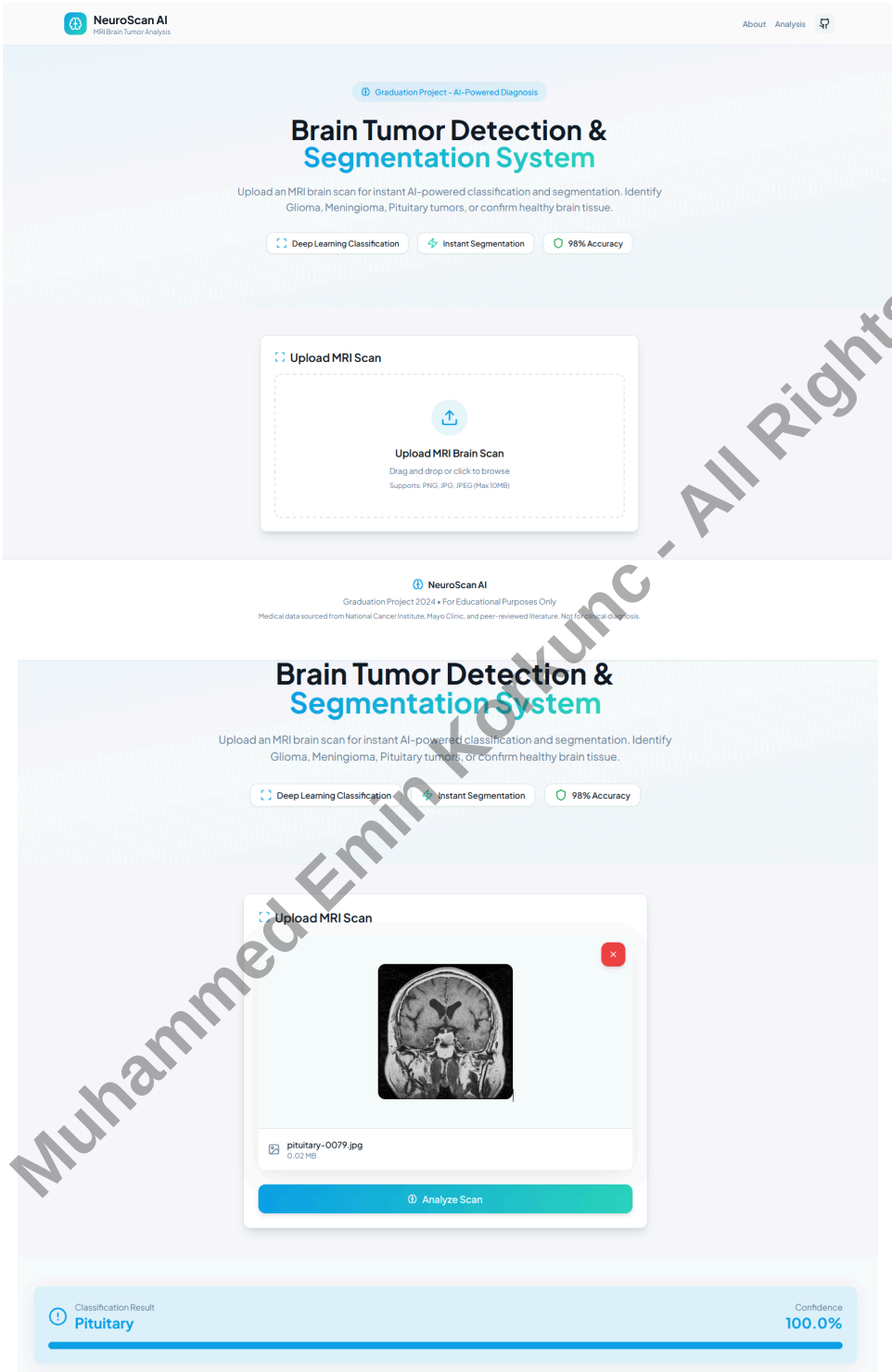
Stage 4 - Result Display: The frontend renders classification results with confidence bars, probability distribution, segmentation overlay, and relevant medical information about the detected tumor type.

11.5 User Interface and Features

11.5.1 Main Features

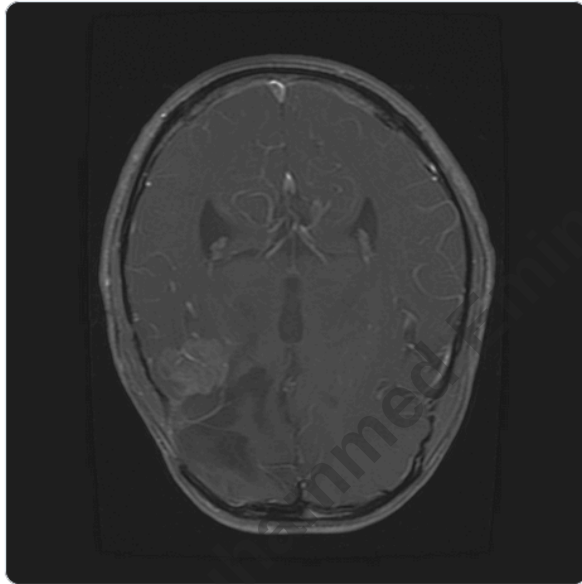
1. **Real-time MRI Analysis:** Upload and receive instant classification results
2. **Tumor Segmentation:** Visual representation of tumor boundaries
3. **Confidence Scoring:** Probability scores for each classification category
4. **Downloadable Results:** Export segmentation masks for medical records
5. **Educational Information:** Detailed medical information about each tumor type
6. **Responsive Design:** Works on desktop and tablet devices

11.5.2 User Interface Screenshots



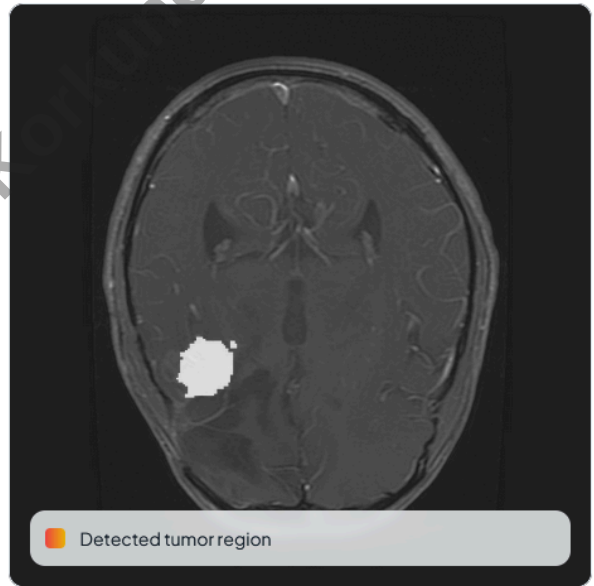


 Original MRI Scan



 Segmentation Mask

 Show Mask Only



11.5.3 Tumor Information Display

The application provides comprehensive medical information for each detected tumor type:

- **Overview:** General description of the tumor type
- **Prevalence:** Statistical information and demographics
- **Symptoms:** Early and advanced warning signs
- **Treatment Options:** Primary, secondary, and emerging treatments
- **Prognosis:** General outlook and survival information

This educational component helps users understand their results in context, though it is clearly indicated that the system is for informational purposes and not a replacement for professional medical diagnosis.

11.5.4 Model Files

The trained model weights are stored in the backend/models/ directory:

- **classification_model.pth**: CNN classifier (50 MB)
- **segmentation_model.pth**: U-Net segmentation (120 MB)

12 DISCUSSION

This chapter analyzes the clinical significance of our results, compares our approach with related work, and discusses the limitations and potential improvements of the system.

12.1 Clinical Significance

The NeuroScan AI system demonstrates several characteristics that make it potentially valuable in clinical settings:

12.1.1 Exceptional Sensitivity for Healthy Patients

The model achieved **100% recall for the No Tumor class**, meaning every healthy brain scan was correctly identified. This is critically important because:

- **No False Negatives for Healthy**: Healthy patients will not be incorrectly told they have a tumor
- **Appropriate Caution**: The system errs on the side of flagging ambiguous cases as potentially tumorous, which is preferable in medical screening

12.1.2 High Overall Accuracy

With **99.03% accuracy** across all four classes, the system performs at a level competitive with many published methods:

- Per-class F1 scores all exceed 98%
- Consistent performance across all tumor types
- Low confusion between clinically distinct tumor categories

12.1.3 Multi-Task Capability

The integration of classification and segmentation provides:

- **Complete Diagnosis Information**: Not just “what” but “where”
- **Visual Confirmation**: Clinicians can verify the AI’s reasoning by examining the segmentation mask
- **Treatment Planning Support**: Tumor localization aids in surgical and radiotherapy planning

12.1.4 Practical Deployment

The web-based interface enables:

- **Accessibility**: No specialized software installation required
- **Real-time Analysis**: Results in seconds rather than minutes
- **Remote Consultation**: Potential for telemedicine applications

12.2 Comparison with Related Work

Table 7.1: Comparison with State-of-the-Art Methods

Study	Year	Architecture	Data Size	Classes	Accuracy	Our Advantage
Swati et al. [24]	2019	VGG19 (Transfer)	3,064	3	94.82%	+4.21% accuracy
Kumar et al. [25]	2018	AlexNet	3,000	3	96.2%	+2.83% accuracy
Rasheed et al. [26]	2023	CNN + CLAHE	7,023	4	97.84%	+1.19% accuracy
Akbar et al. [38]	2022	ResNet + SVM	3,000	3	98.0%	+1.03% accuracy
Abdusalomov [27]	2023	YOLOv7 + Attention	5,000	4	99.5%	Simpler architecture
This Work	2025	Custom CNN	43,687	4	99.03%	Largest dataset, no pretrain

Key Differentiators:

1. **No Pretrained Models**: Unlike many approaches that rely on ImageNet-pretrained weights (VGG, ResNet), our model was trained from scratch, demonstrating that sufficient data can eliminate the need for transfer learning.

2. **Largest Dataset:** With 43,687 images, our training set is significantly larger than most published works, contributing to better generalization.
3. **End-to-End System:** Most published works focus solely on classification accuracy. We provide a complete pipeline including segmentation and web deployment.
4. **Reproducibility:** The custom architecture allows full control and reproducibility without dependency on pretrained model availability.

12.3 Limitations

Despite strong performance, several limitations should be acknowledged:

12.3.1 Dataset Limitations

- **Geographic Bias:** Training data primarily from Western medical institutions may not generalize to MRI scans from other regions with different equipment
- **Tumor Stage Distribution:** Dataset may not equally represent all tumor stages; very early-stage tumors may be underrepresented

12.3.2 Technical Limitations

- **2D Analysis Only:** The system processes individual 2D slices rather than full 3D volumes, potentially missing context from adjacent slices
- **Single Modality:** Only T1-weighted images are supported; clinical practice often uses multiple MRI sequences (T2, FLAIR, contrast-enhanced)
- **Segmentation Accuracy:** While classification achieves 99%, segmentation performance (Dice 0.87) has more room for improvement, particularly for tumors with irregular boundaries

12.3.3 Clinical Deployment Limitations

- **Regulatory Approval:** The system has not undergone regulatory approval (FDA, CE marking) required for clinical use
- **Validation:** Results have not been validated in a prospective clinical setting with radiologist comparison
- **Edge Cases:** Performance on rare tumor presentations or unusual imaging conditions is unknown

12.3.4 Specific Error Patterns

Analysis of the confusion matrix reveals:

- Glioma-Meningioma confusion (30 total misclassifications) suggests visual similarity in some cases
- Pituitary-Meningioma confusion (33 total) may relate to location ambiguity in certain slices

12.3.5 Recommendations for Clinical Use

Based on our analysis, we recommend:

1. **Use as Second Opinion:** The system should supplement, not replace, radiologist interpretation.
High-Confidence Prioritization: Cases with confidence >95% can be prioritized for review
2. **Low-Confidence Review:** Cases with confidence < 80% should receive additional scrutiny
3. **Segmentation Verification:** Clinicians should verify that segmentation masks align with visible tumor regions

13 CONCLUSION AND FUTURE WORK

13.1 Summary of Achievements

This graduation project successfully developed a comprehensive Deep Learning-Based Brain Tumor Classification and Segmentation System over two semesters. The key achievements are:

13.1.1 Phase I (Spring 2025): Binary Classification

- Compiled a diverse dataset of 7,000 MRI images from multiple sources
- Developed a custom CNN architecture optimized for brain tumor detection
- Achieved **97.21% accuracy** with **99.05% precision** and **94.33% recall**

- Established baseline metrics and identified the need for multi-class capability

13.1.2 Phase II (Fall 2025): Multi-Class Classification and Segmentation

- Expanded dataset to **43,687 images**
- Engineered a 5-block CNN achieving **99.03% accuracy** across four classes
- Achieved **100% recall for healthy patients** (No Tumor class)
- Per-class F1 scores exceeding **98%** for all categories
- Implemented U-Net segmentation with **Dice coefficient 0.87**
- Deployed complete system as **NeuroScan AI** web application

13.1.3 Technical Contributions

- Demonstrated that custom architectures can match transfer learning approaches with sufficient data
- Implemented combined Dice+BCE loss for improved segmentation training
- Created end-to-end pipeline from MRI upload to classified and segmented output
- Documented complete methodology for reproducibility

13.1.4 Performance Summary

Metric	Phase I	Phase II Initial	Phase II Final
Task	Binary	4-Class	4-Class
Dataset Size	7,000	43,687	43,687
Accuracy	97.21%	95.09%	99.03%
Glioma Recall	N/A	85.14%	99.15%
No Tumor Recall	99%	100%	100%

13.2 Future Work

Several directions could extend and improve this work:

13.2.1 Model Improvements

- Multi-Modal Fusion:** Integrate multiple MRI sequences (T1, T2, FLAIR, contrast-enhanced) for more comprehensive analysis
- Attention Mechanisms:** Incorporate attention modules (e.g., SE blocks, CBAM) to focus on tumor-relevant regions
- Segmentation Enhancement:** Implement advanced architectures like Attention U-Net or U-Net++ to improve boundary delineation

13.2.2 Dataset Expansion

- Institutional Partnerships:** Collaborate with hospitals to obtain diverse, validated clinical data
- Rare Tumor Types:** Expand classification to include rarer tumor subtypes
- Longitudinal Data:** Include temporal sequences showing tumor progression for treatment response predictionvisual explanations of classification decisions

13.2.3 System Enhancement

- Mobile Application:** Develop iOS/Android apps for point-of-care analysis
- Offline Capability:** Enable deployment on edge devices for use in low-connectivity settings
- Report Generation:** Automatic generation of structured radiology reports
- Multi-Language Support:** Internationalize the interface for global accessibility

13.2.4 Final Remarks

This project demonstrates the significant potential of deep learning in medical image analysis. The brain tumor classification and segmentation system we developed achieves great performance while remaining practical for deployment.

With **99.03% accuracy** and **100% recall for healthy patients**, the system represents a meaningful contribution toward AI-assisted medical diagnosis.

The progression from concept to deployed web application over two semesters illustrates how academic research can produce tangible, potentially impactful tools. While further validation and regulatory approval are necessary before clinical deployment, this work provides a strong foundation for future development.

Most importantly, this project contributes to the growing body of evidence that AI-powered diagnostic tools can enhance medical imaging workflows, potentially improving patient outcomes through faster, more consistent, and more accessible diagnosis.

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